

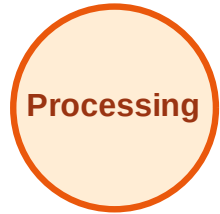
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



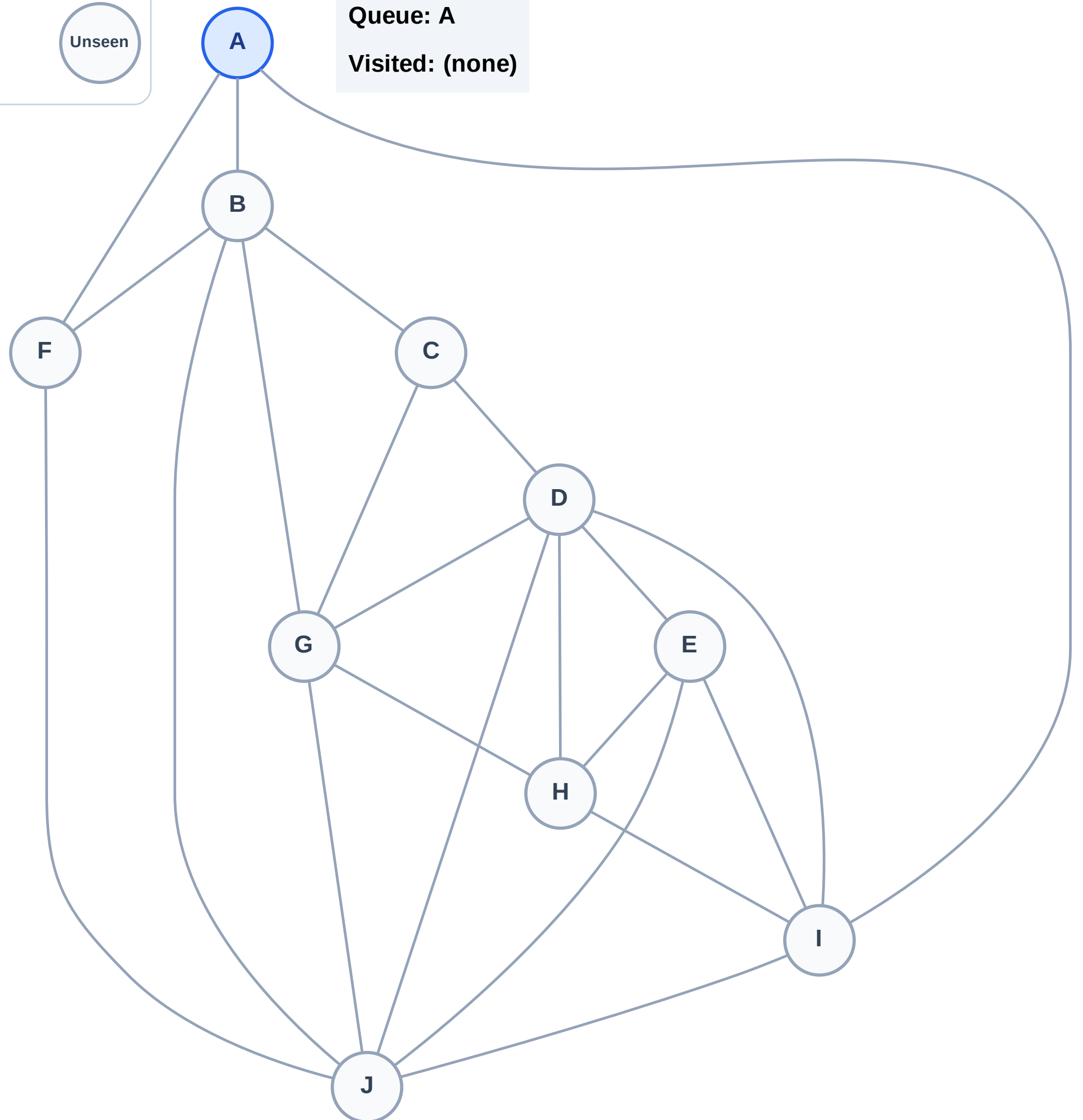
Discovered



Unseen

Queue: A

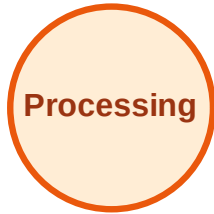
Visited: (none)



BFS Traversal

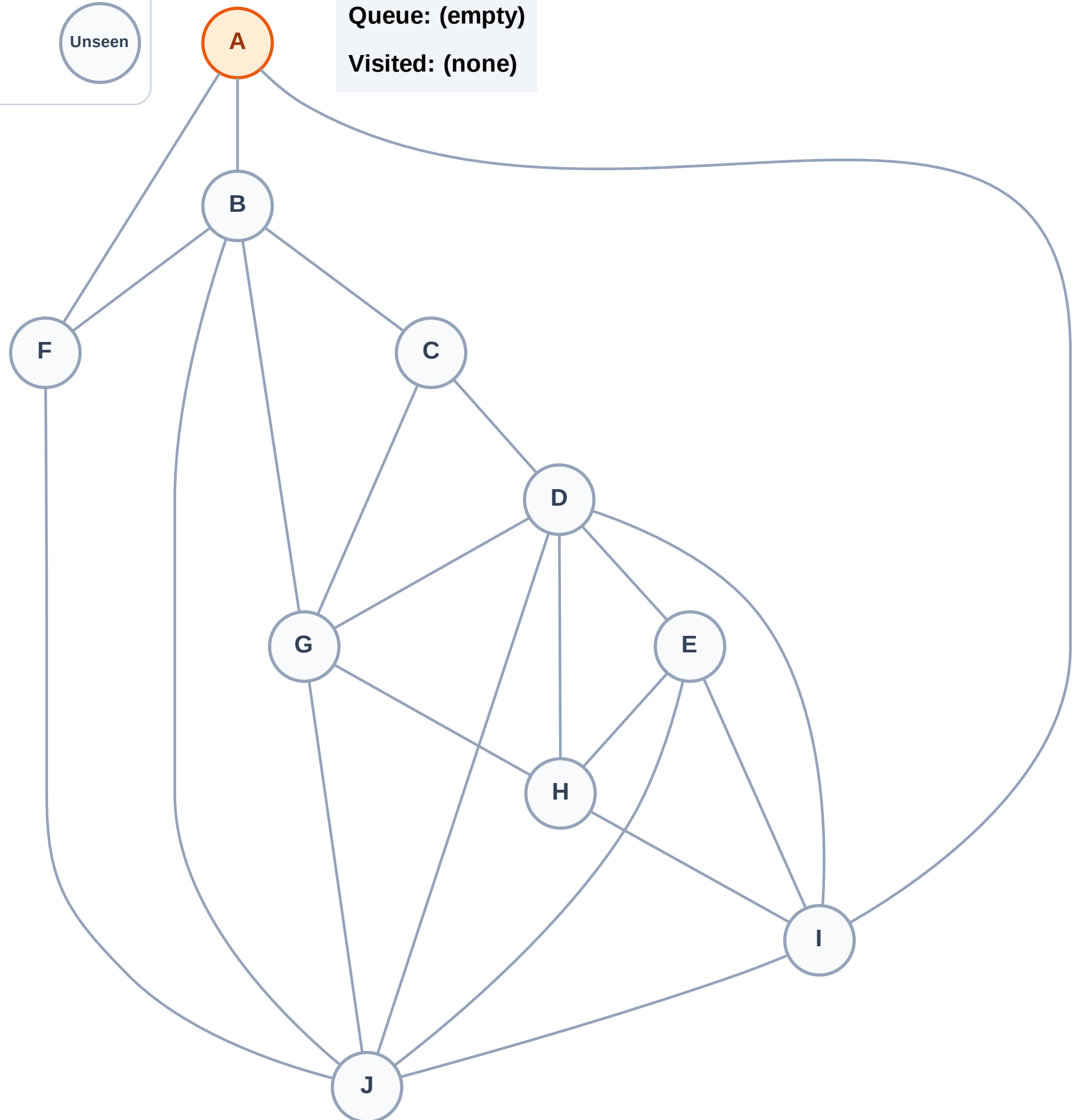
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

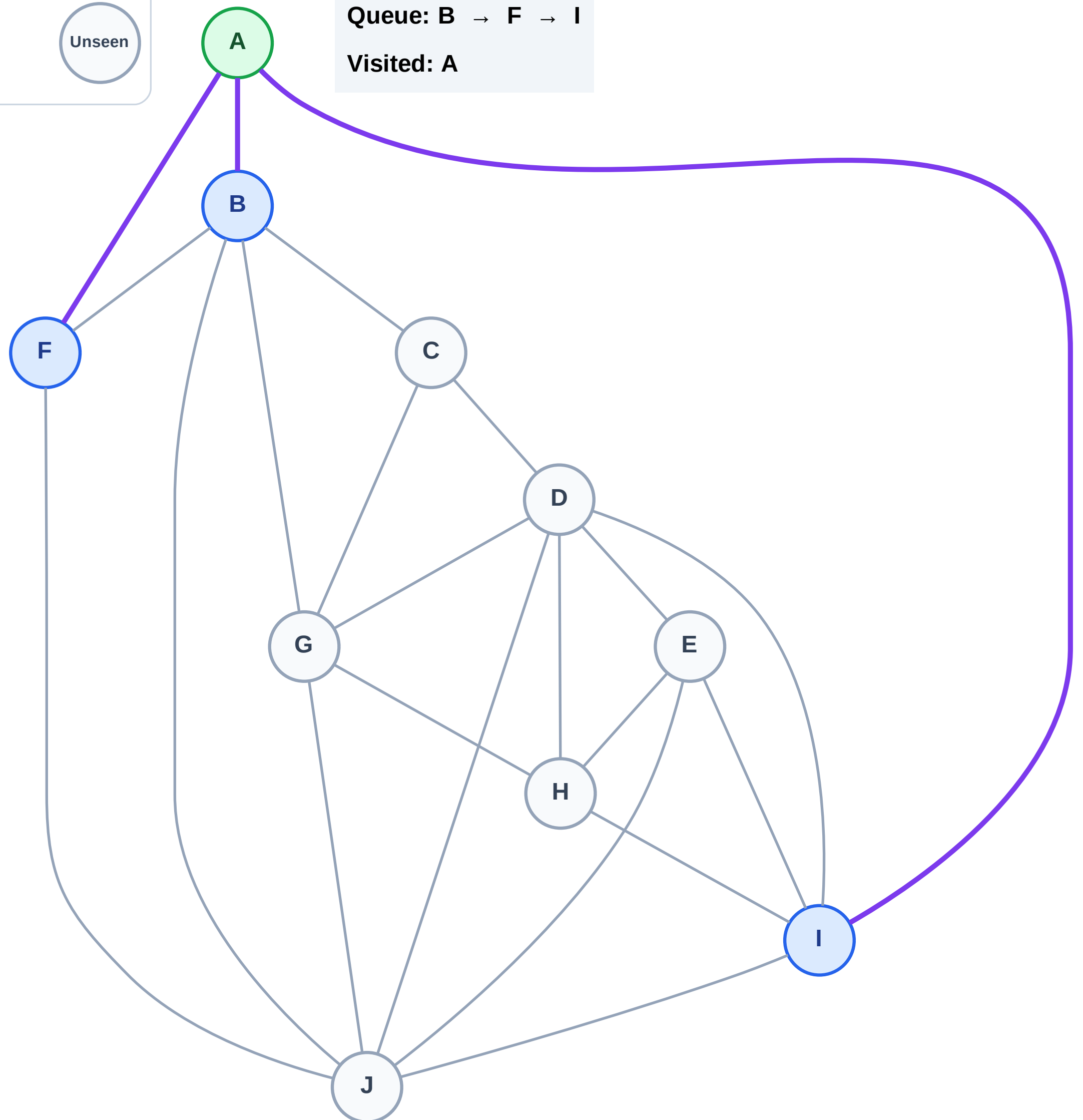
Step 3: Discover B, F, I from A

Legend



Queue: B → F → I

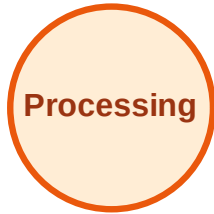
Visited: A



BFS Traversal

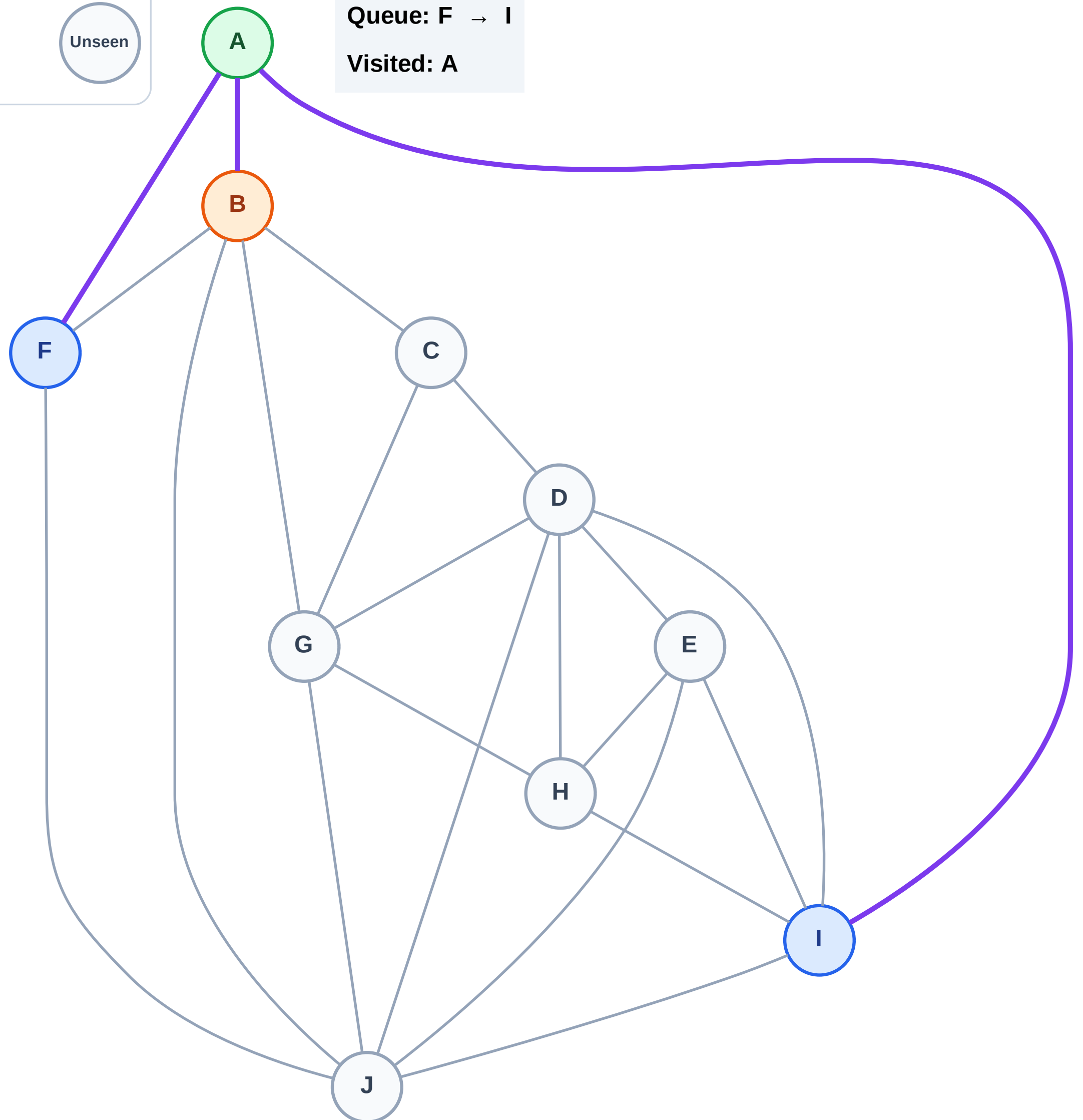
Step 4: Process node B

Legend



Queue: F → I

Visited: A



BFS Traversal

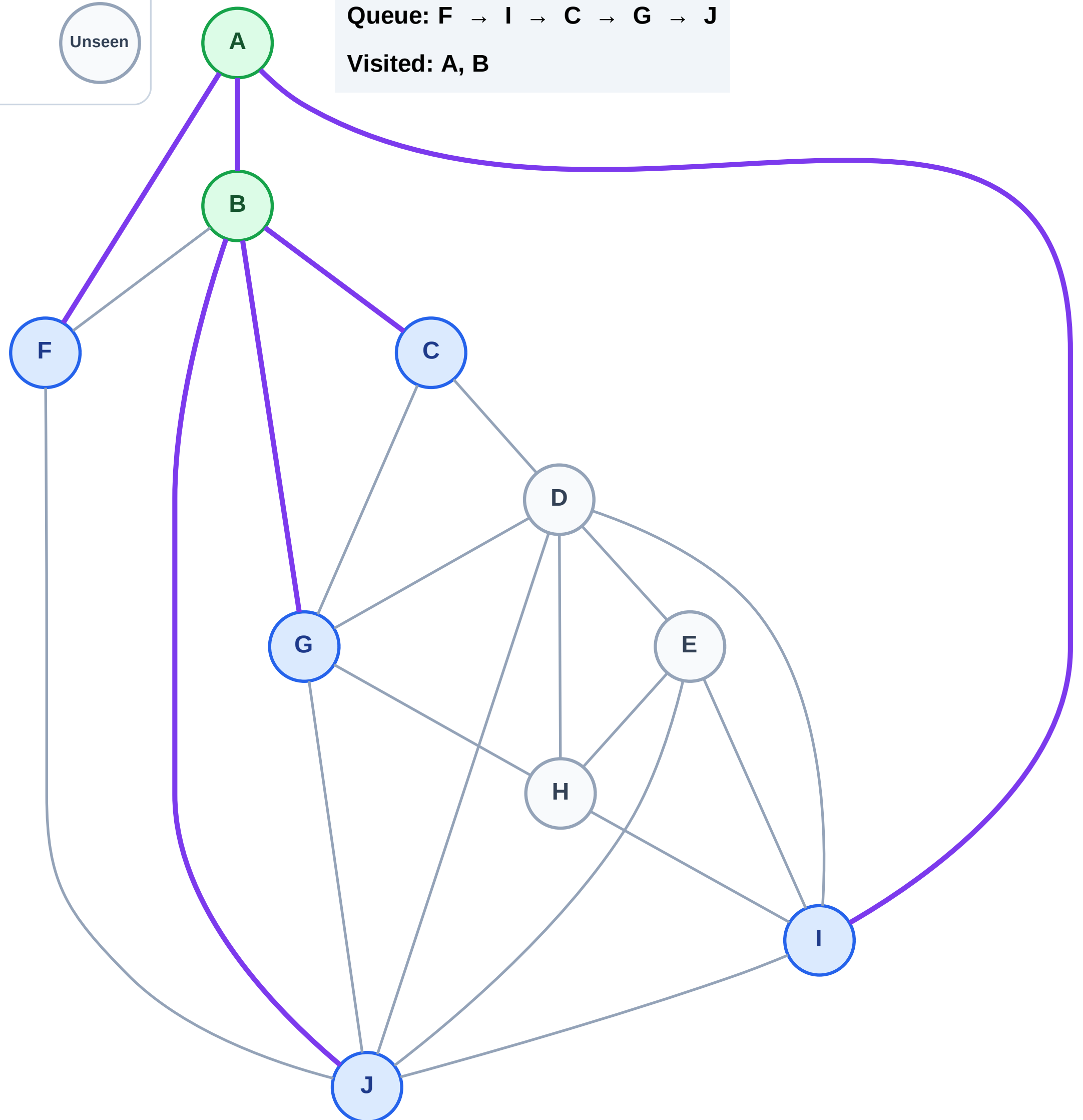
Step 5: Discover C, G, J from B

Legend



Queue: F → I → C → G → J

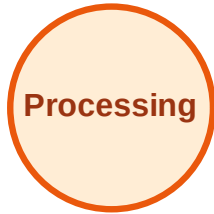
Visited: A, B



BFS Traversal

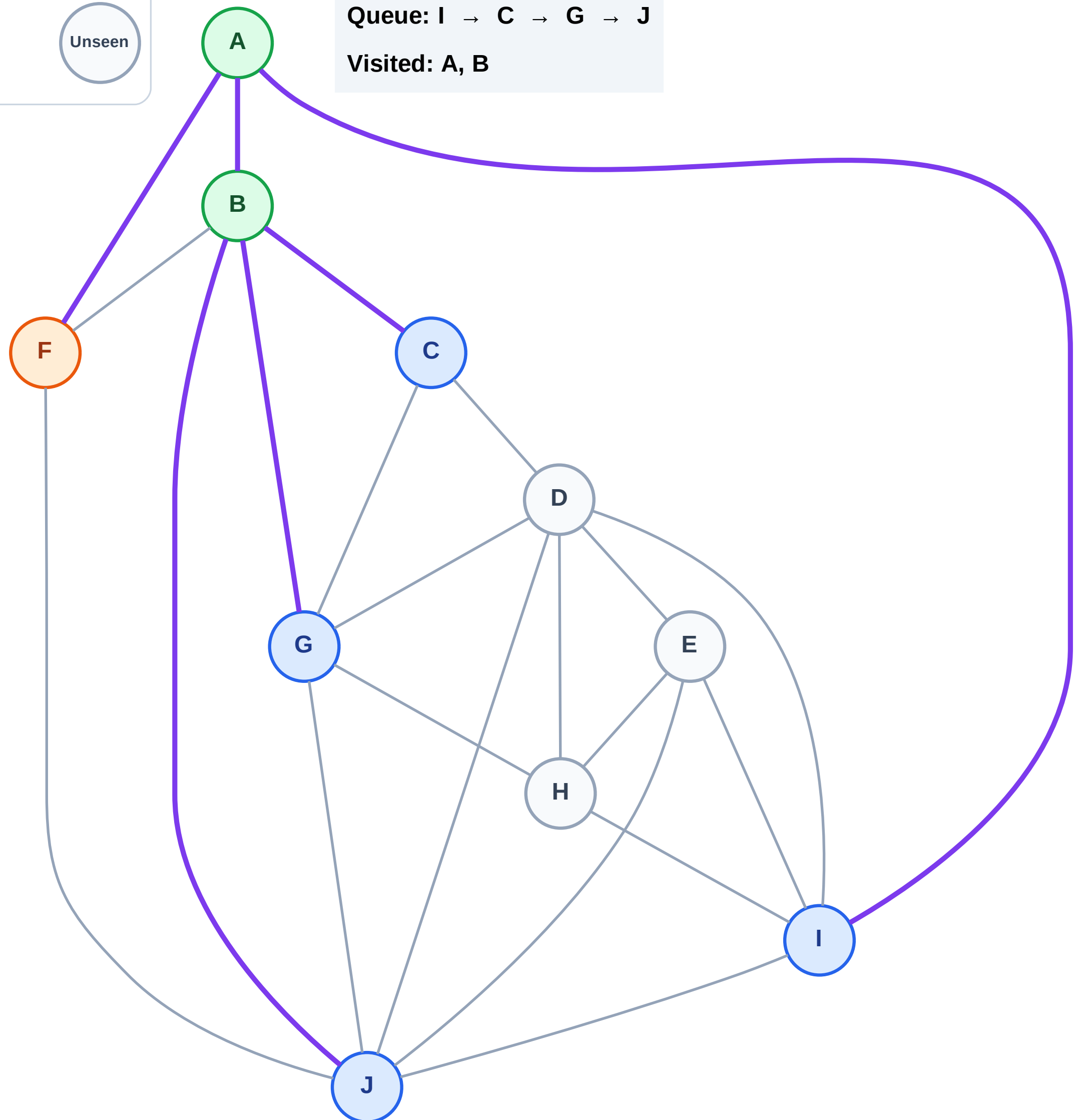
Step 6: Process node F

Legend



Queue: I → C → G → J

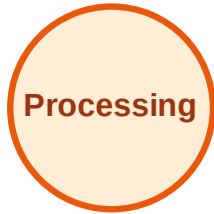
Visited: A, B



BFS Traversal

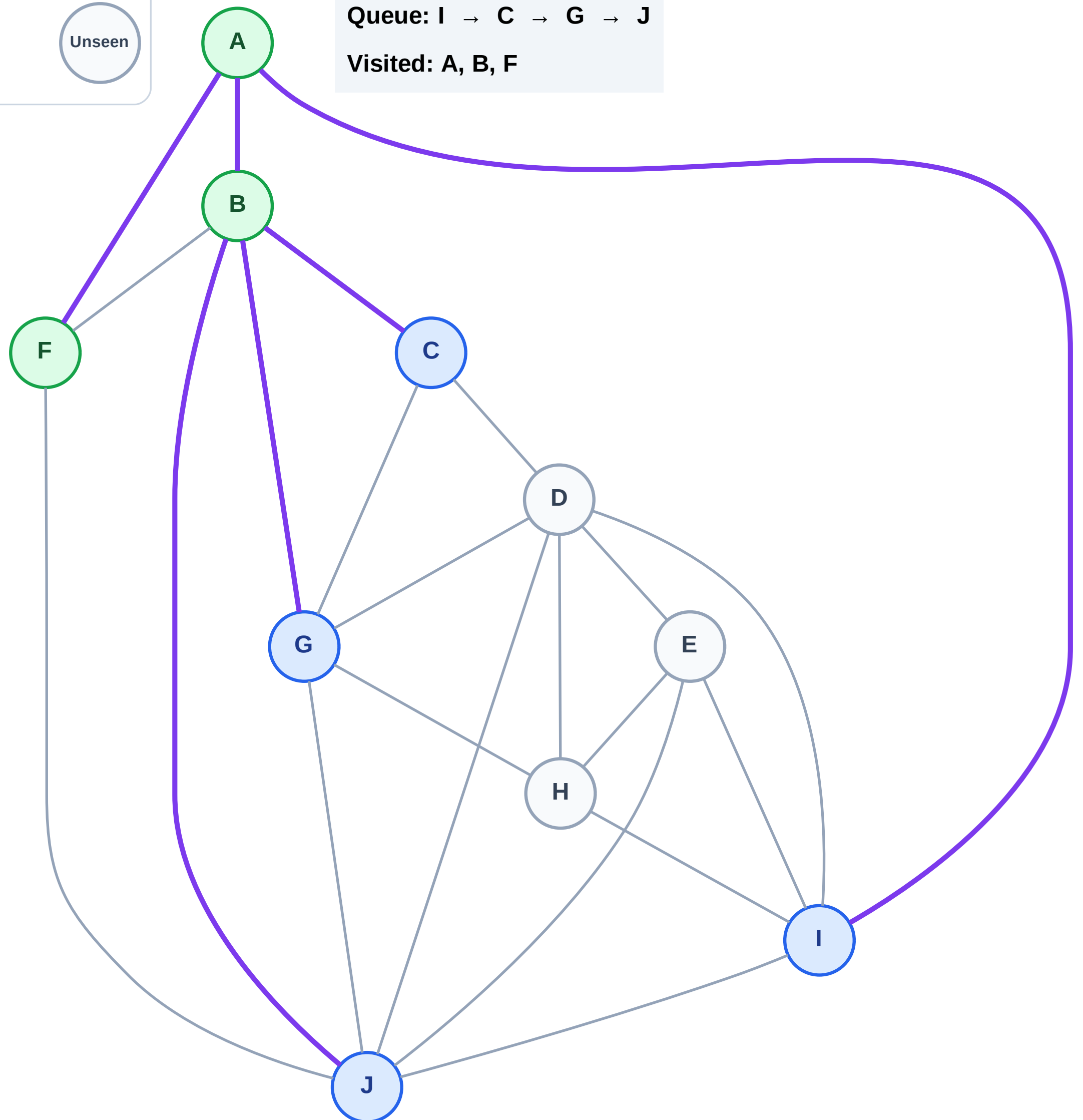
Step 7: No new neighbors from F

Legend



Queue: I → C → G → J

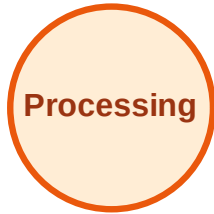
Visited: A, B, F



BFS Traversal

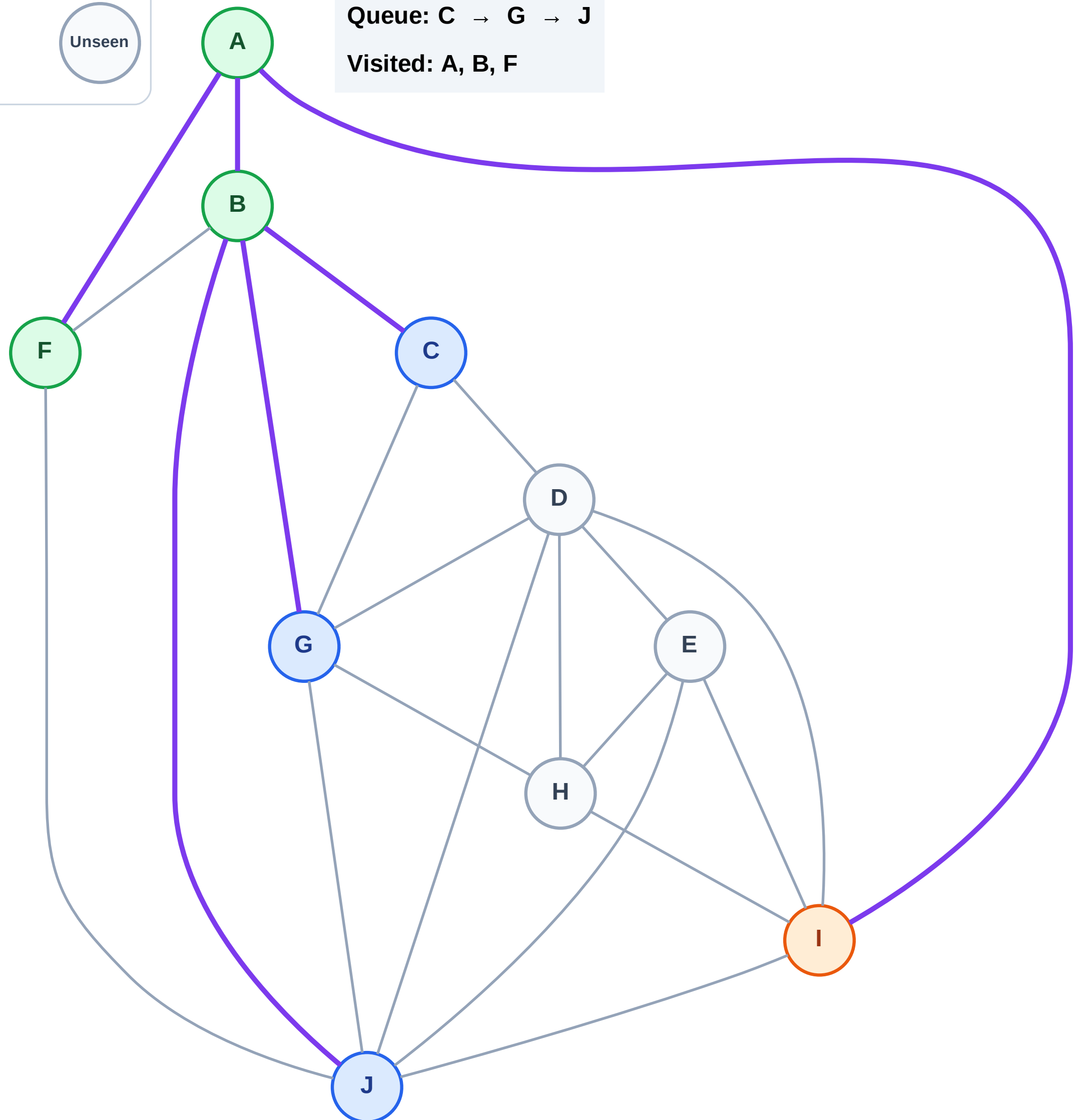
Step 8: Process node I

Legend



Queue: C → G → J

Visited: A, B, F

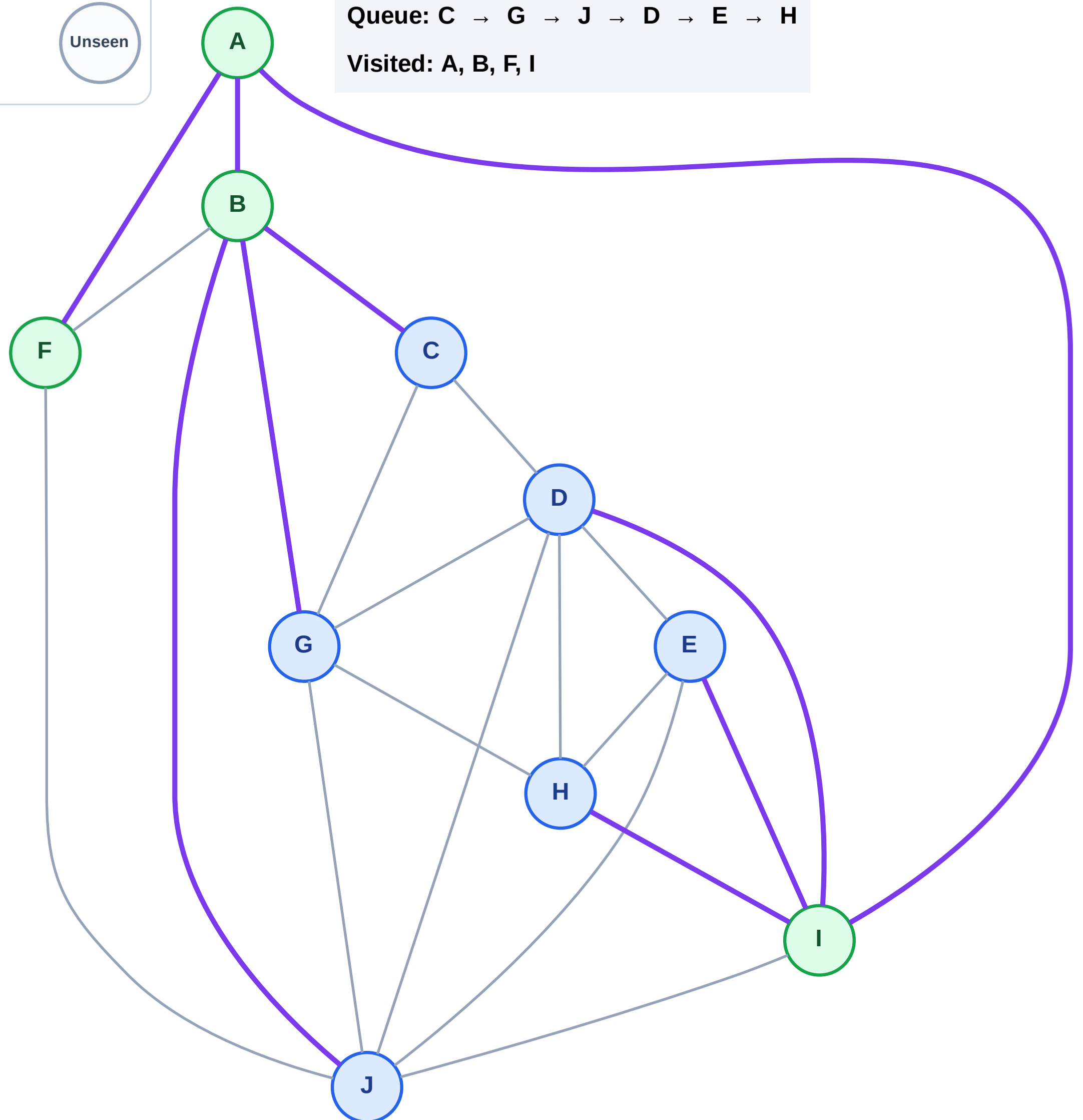


Step 9: Discover D, E, H from I

Queue: C → G → J → D → E → H

Legend

Unseen



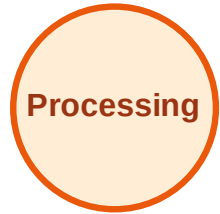
BFS Traversal

Step 10: Process node C

Legend



Complete



Processing



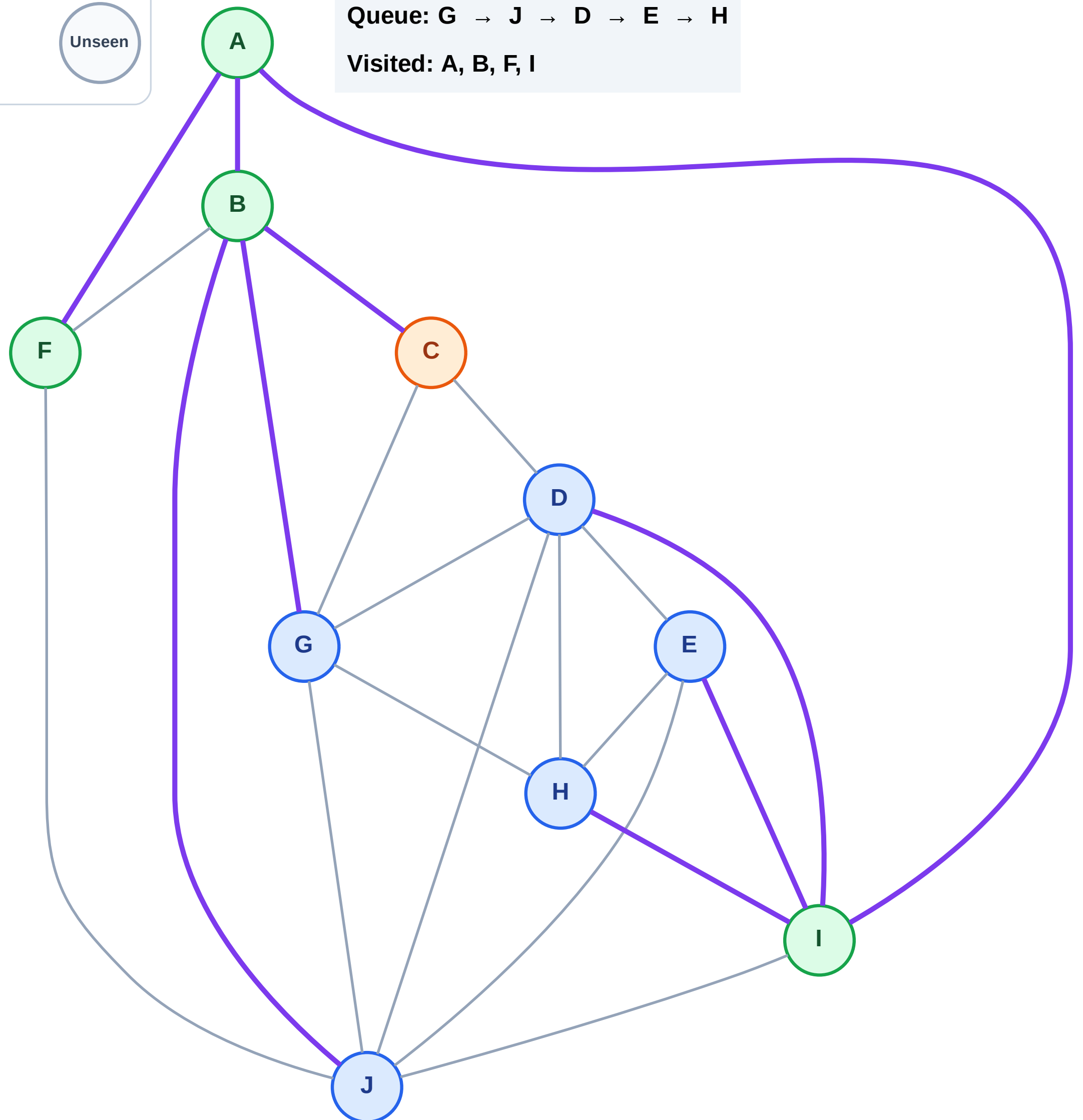
Discovered



Unseen

Queue: G → J → D → E → H

Visited: A, B, F, I



BFS Traversal

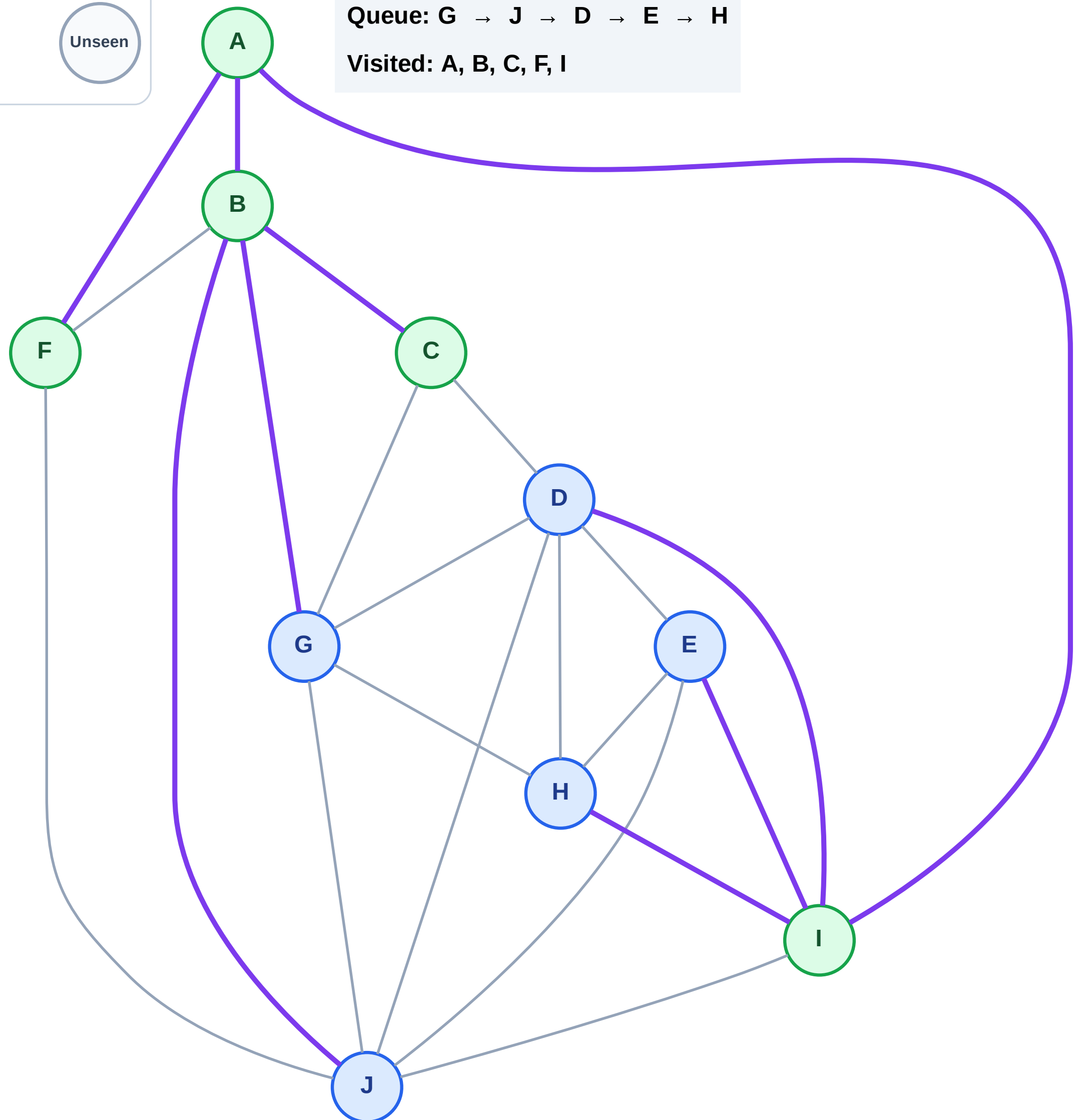
Step 11: No new neighbors from C

Legend



Queue: G → J → D → E → H

Visited: A, B, C, F, I



BFS Traversal

Step 12: Process node G

Legend

Complete

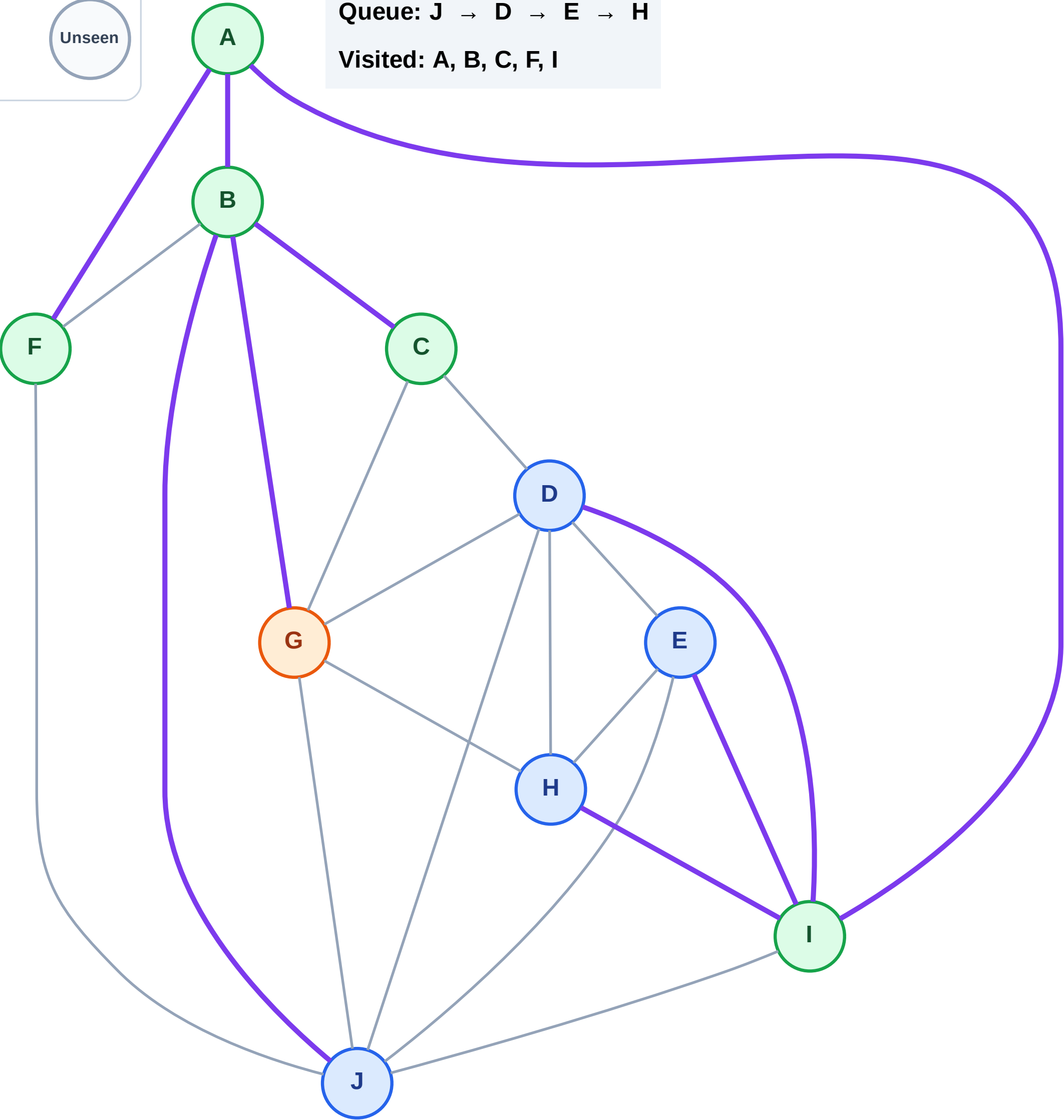
Processing

Discovered

Unseen

Queue: J → D → E → H

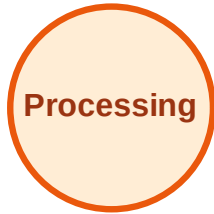
Visited: A, B, C, F, I



BFS Traversal

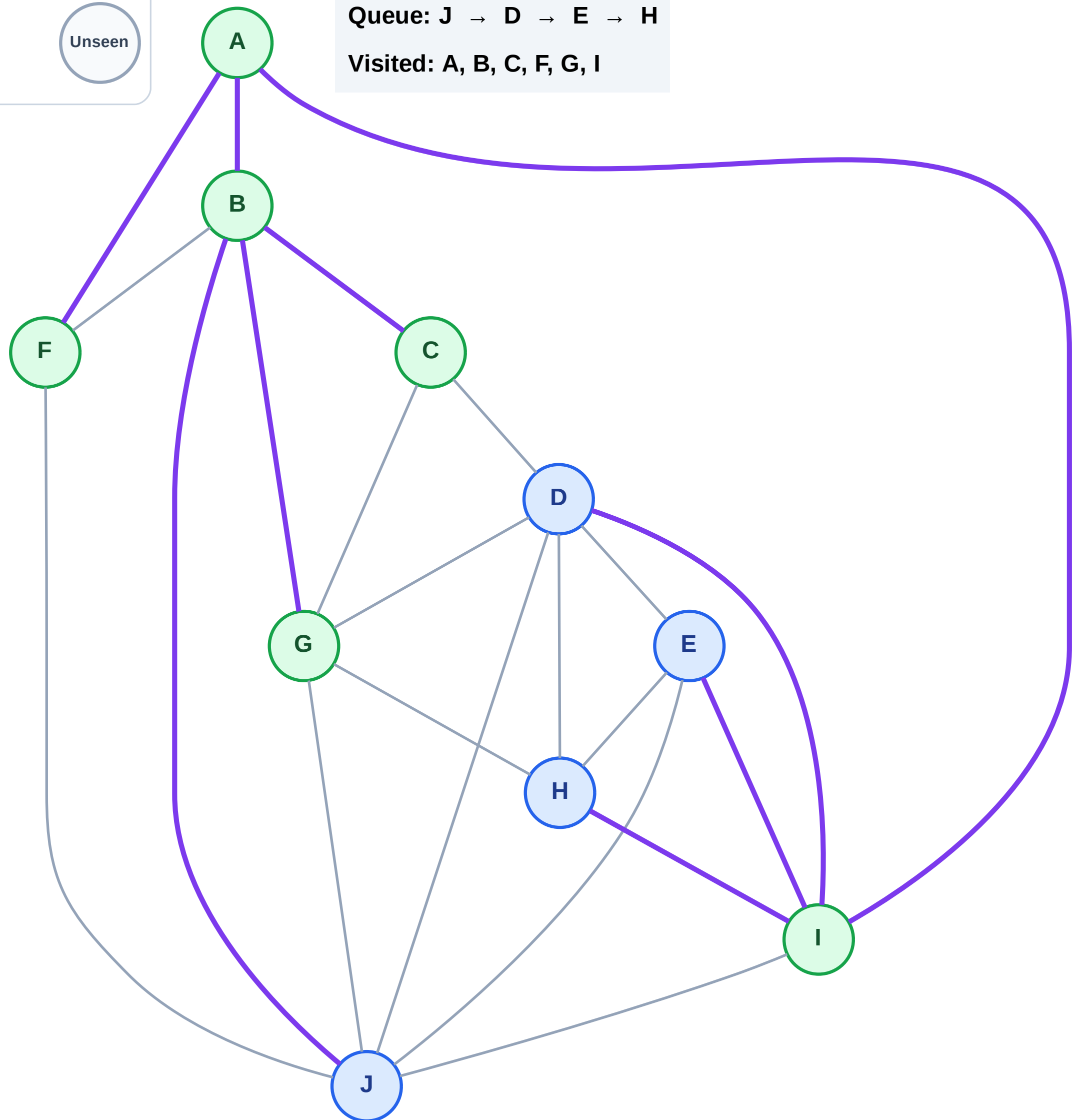
Step 13: No new neighbors from G

Legend



Queue: J → D → E → H

Visited: A, B, C, F, G, I



BFS Traversal

Step 14: Process node J

Legend



Complete



Processing



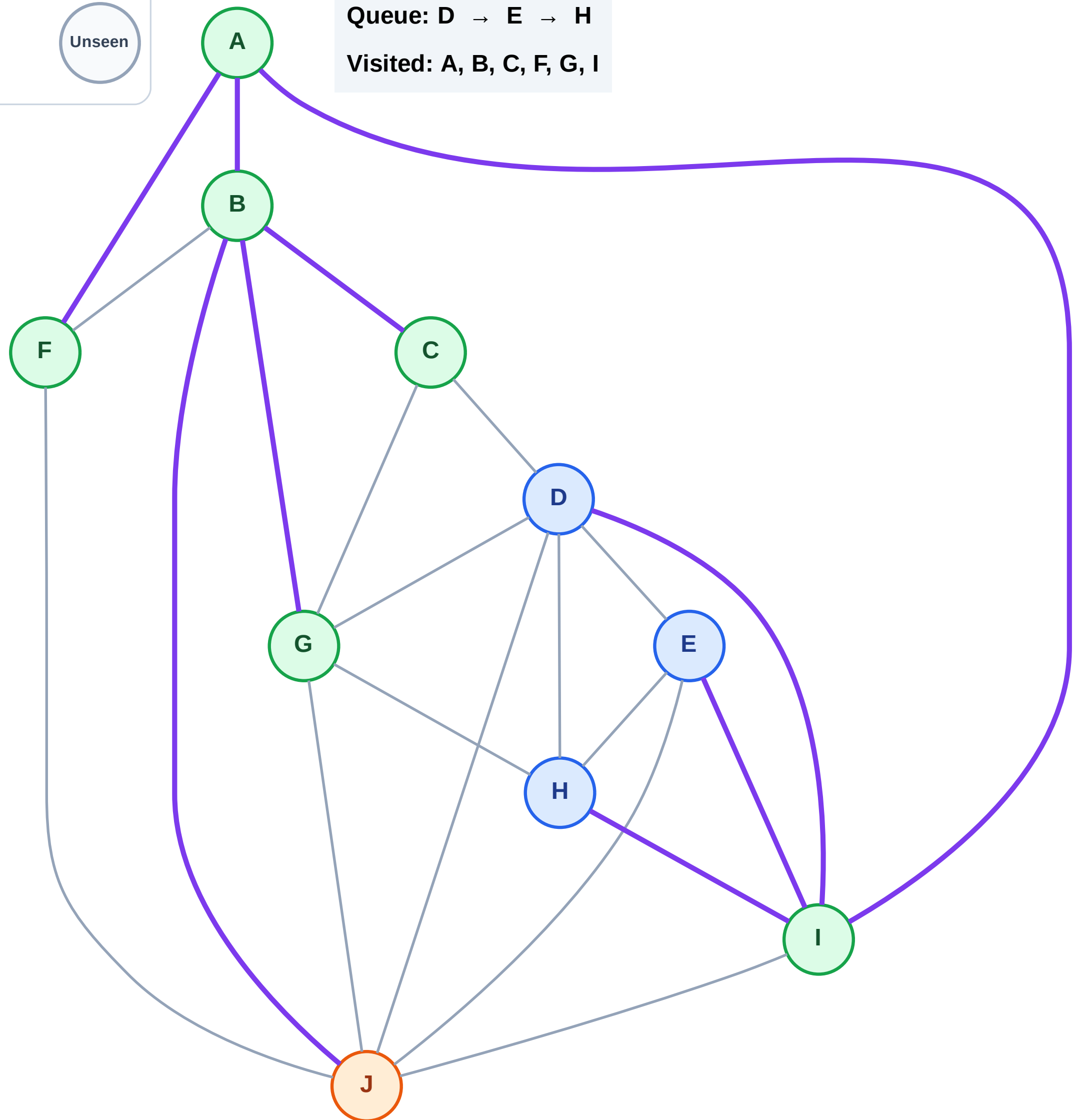
Discovered



Unseen

Queue: D → E → H

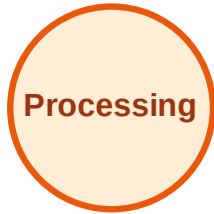
Visited: A, B, C, F, G, I



BFS Traversal

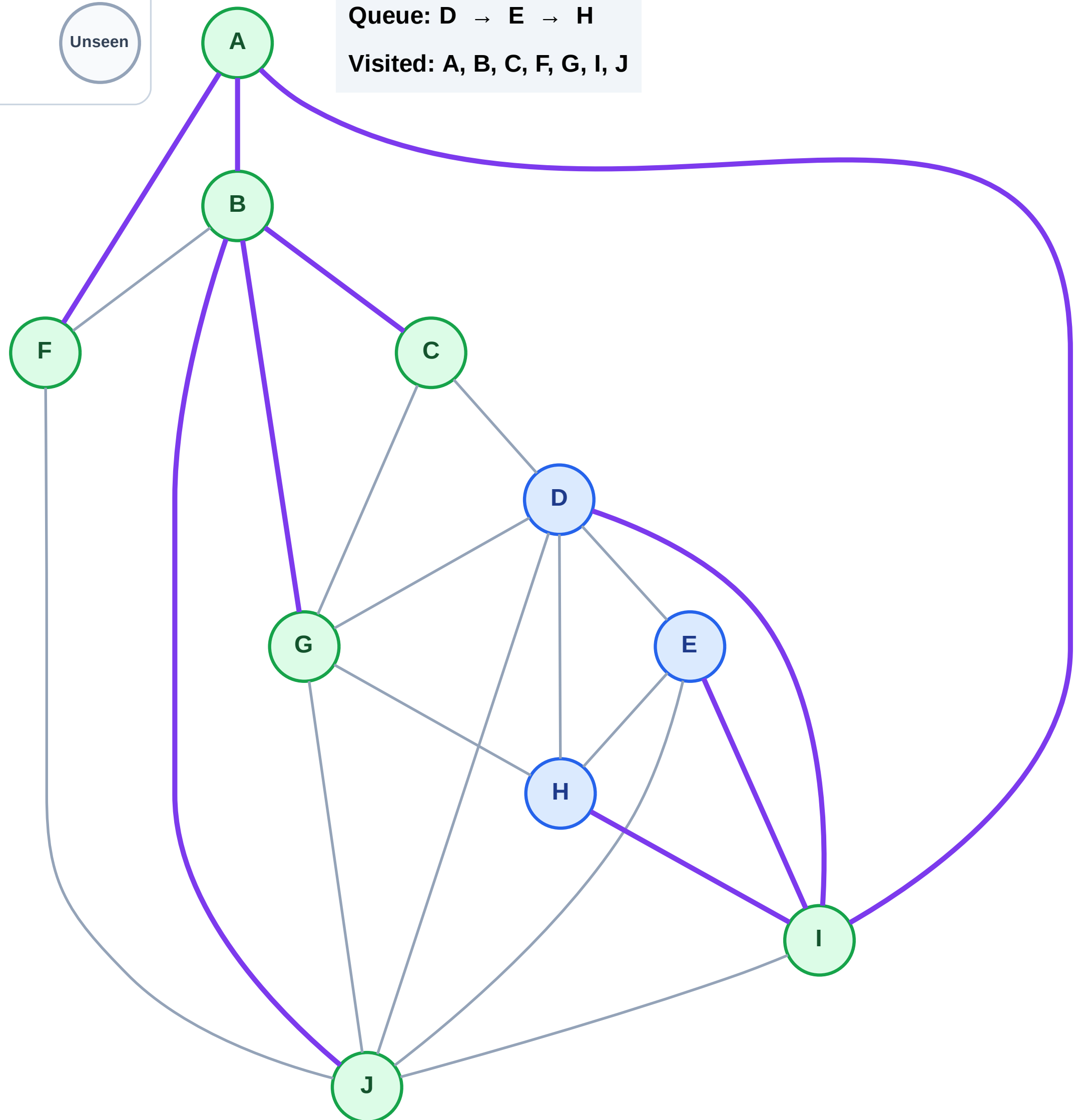
Step 15: No new neighbors from J

Legend



Queue: D → E → H

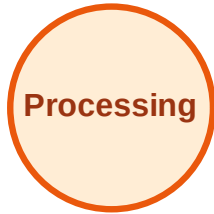
Visited: A, B, C, F, G, I, J



BFS Traversal

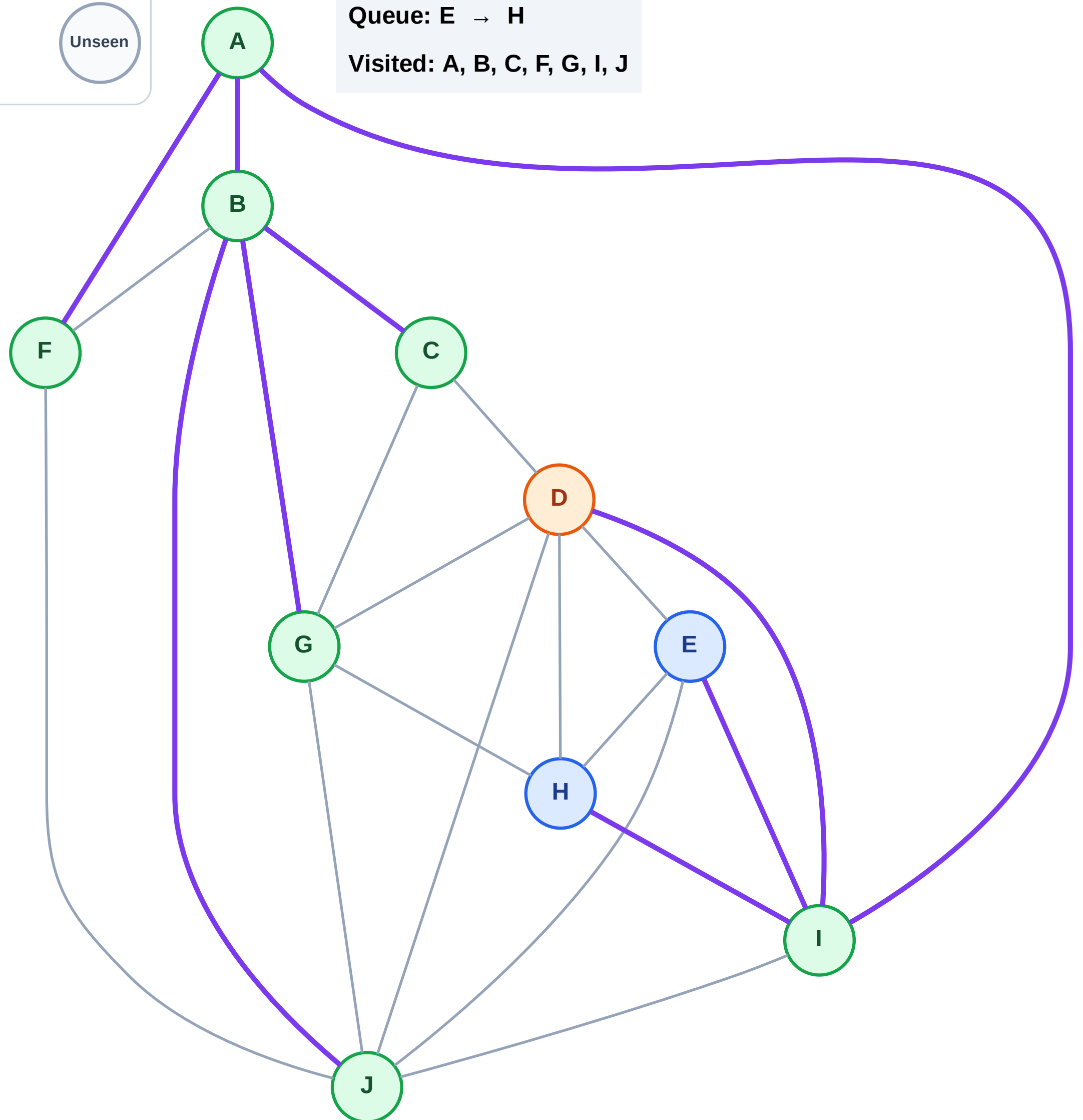
Step 16: Process node D

Legend



Queue: E → H

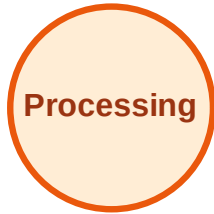
Visited: A, B, C, F, G, I, J



BFS Traversal

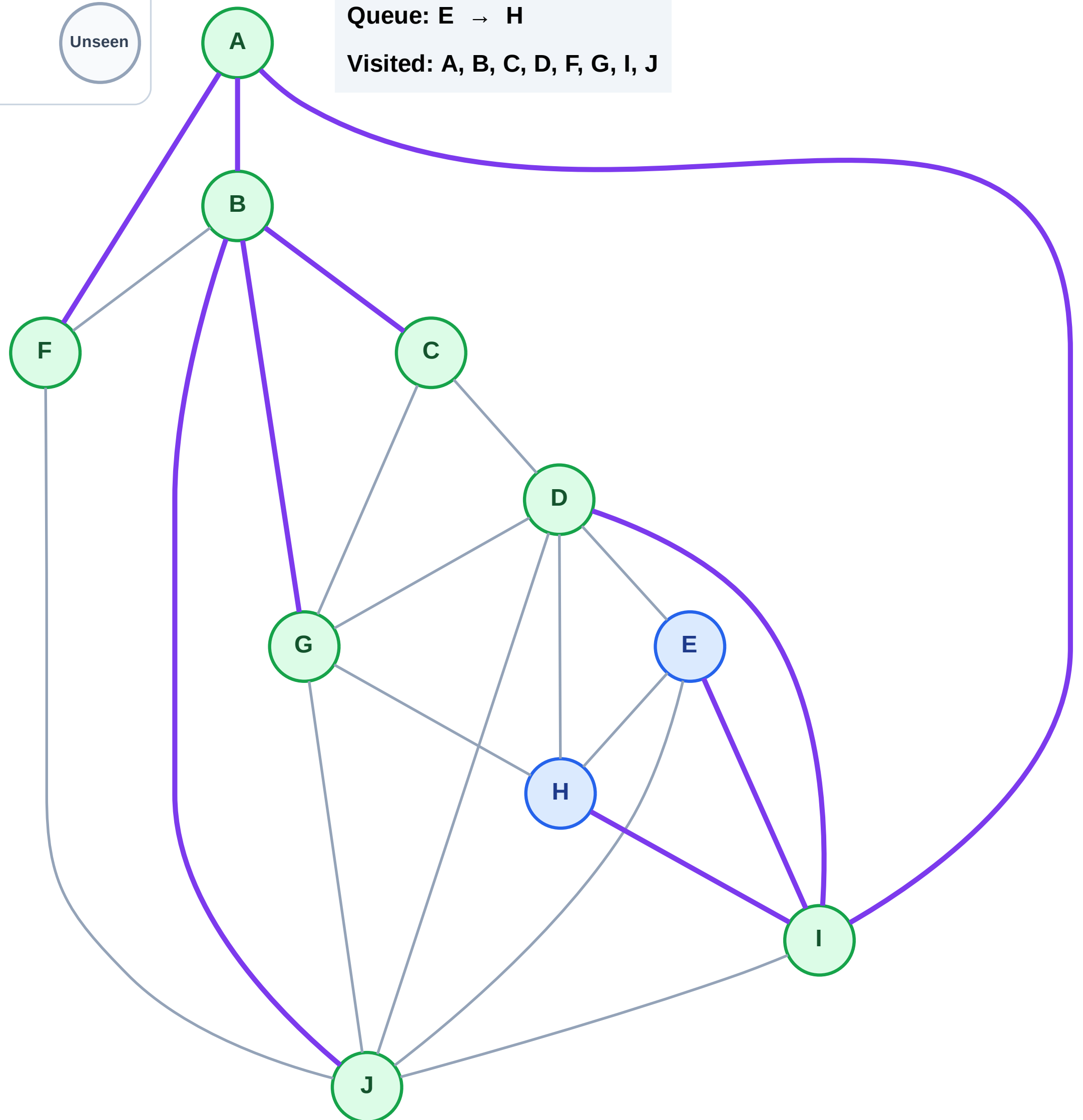
Step 17: No new neighbors from D

Legend



Queue: E → H

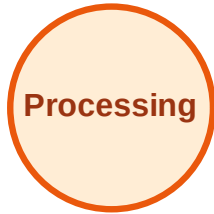
Visited: A, B, C, D, F, G, I, J



BFS Traversal

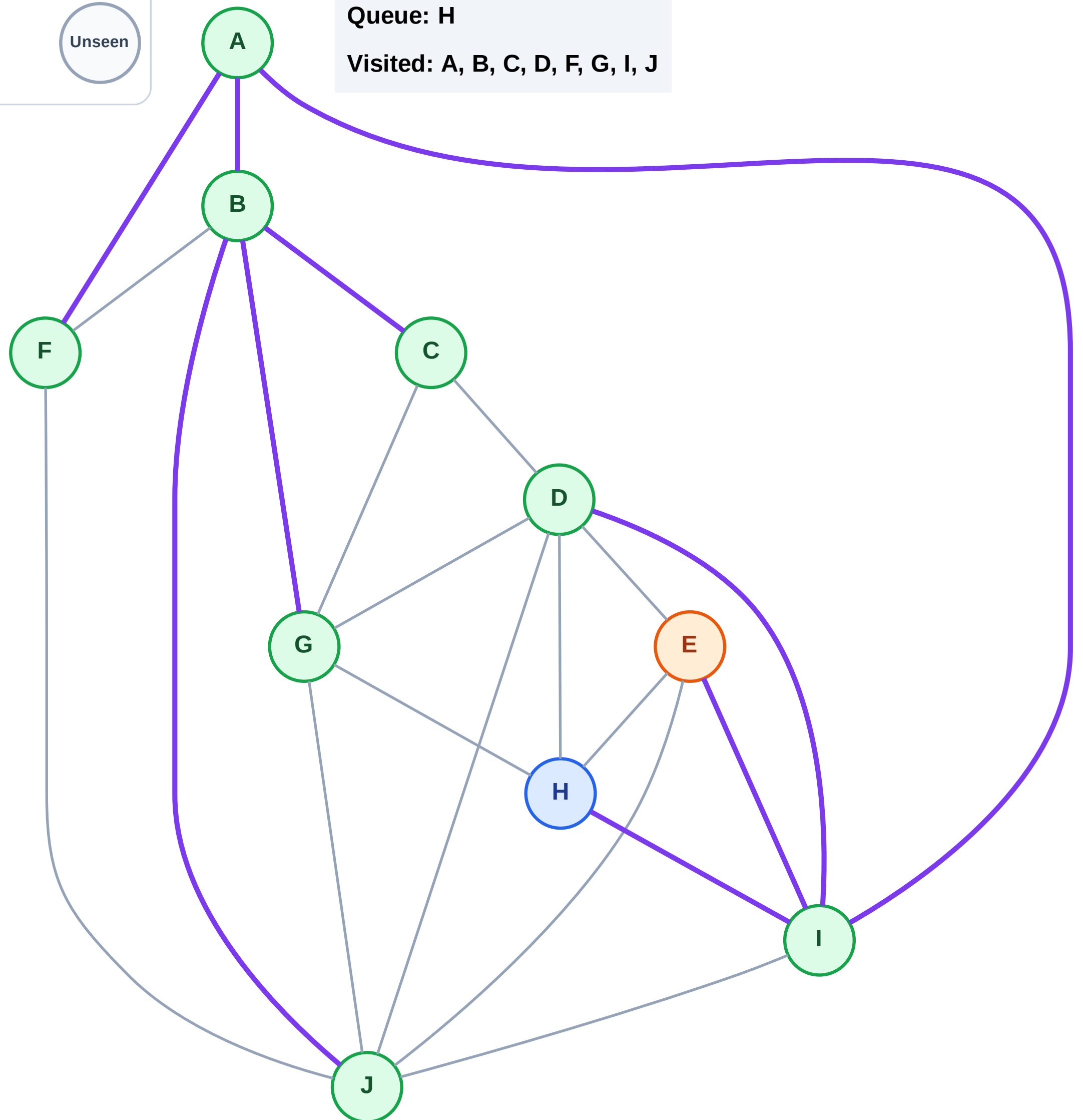
Step 18: Process node E

Legend



Queue: H

Visited: A, B, C, D, F, G, I, J



BFS Traversal

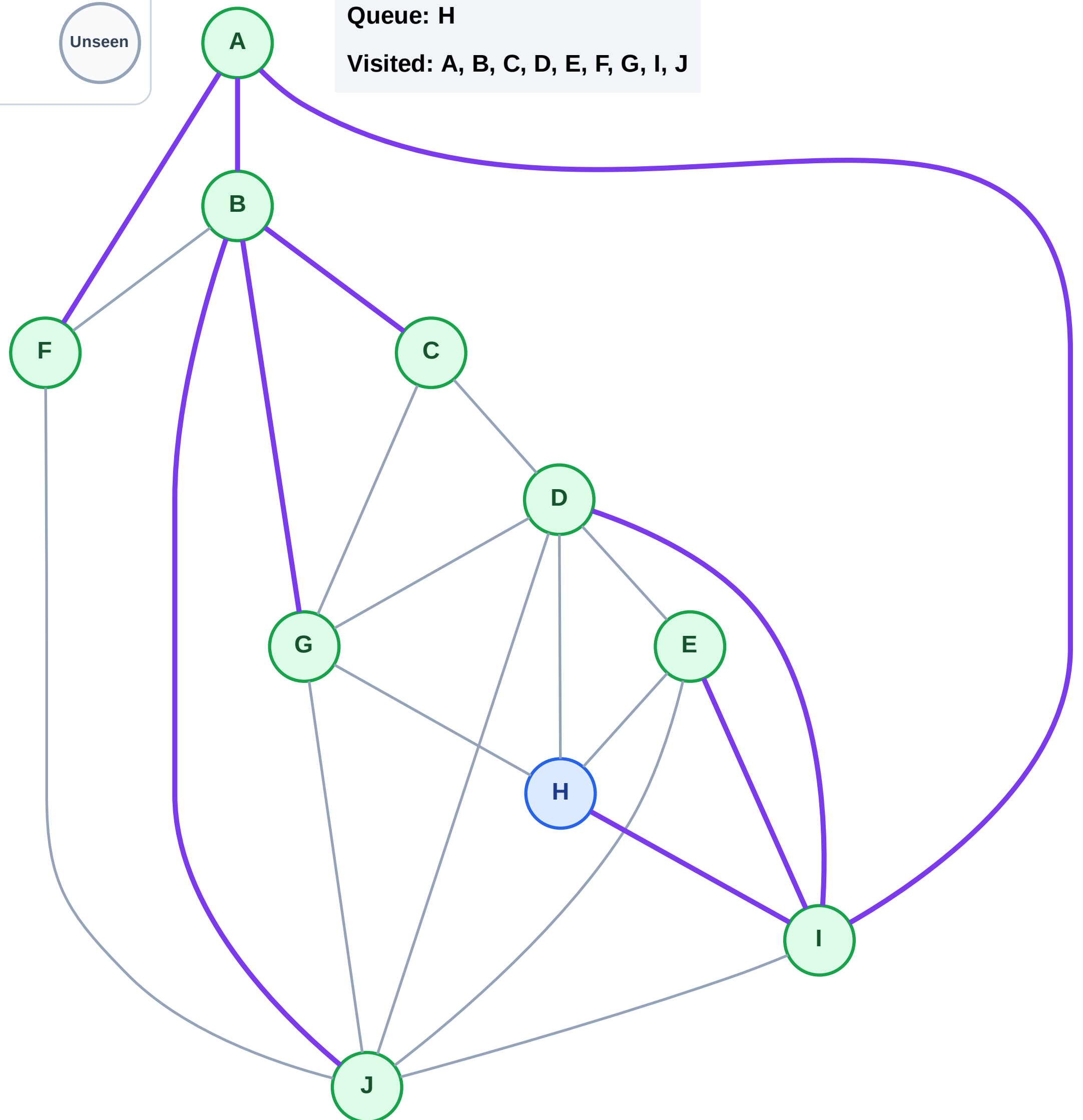
Step 19: No new neighbors from E

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J

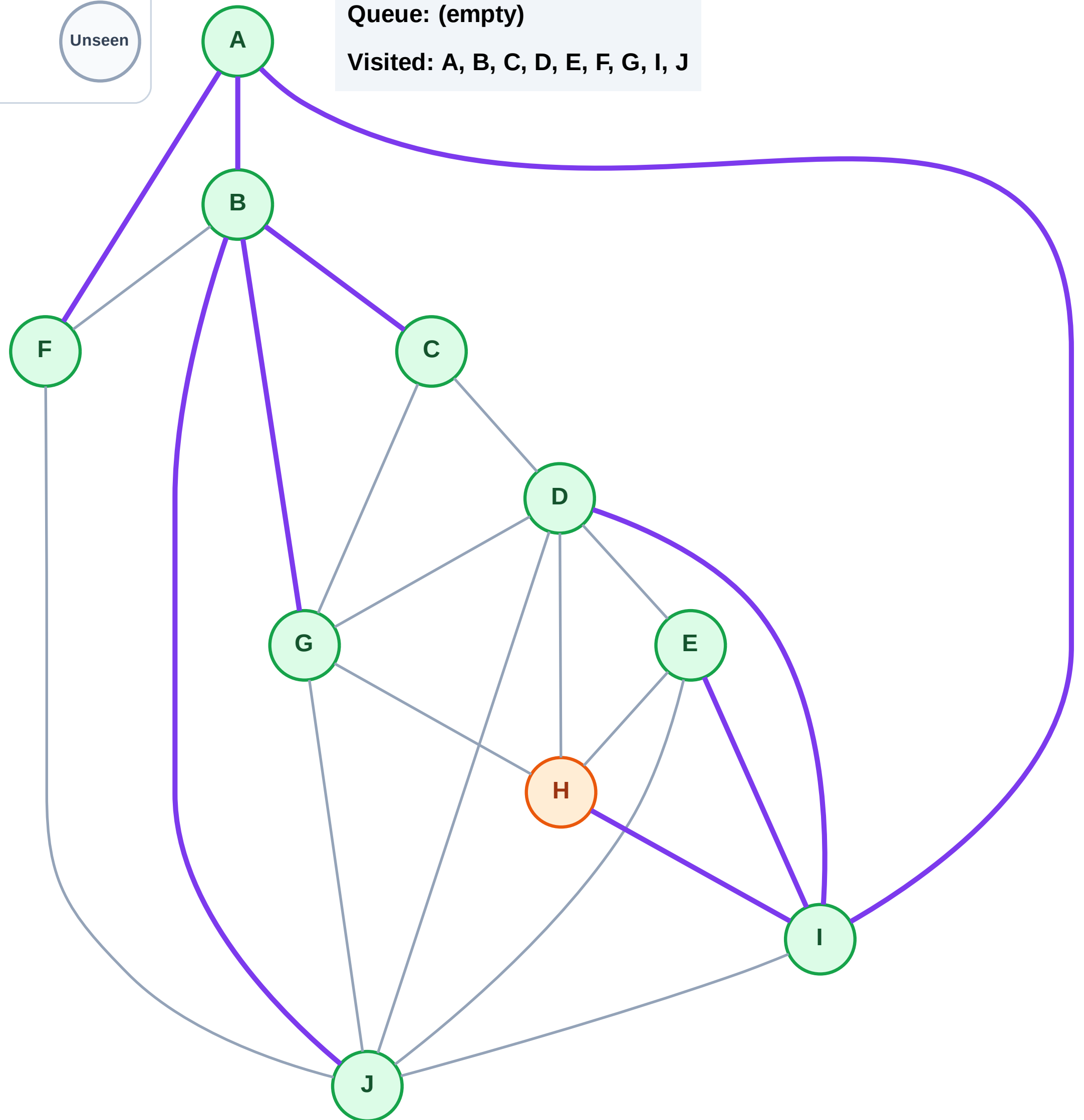


Step 20: Process node H

Queue: (empty)

Legend

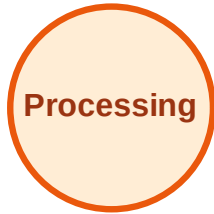
Unseen



BFS Traversal

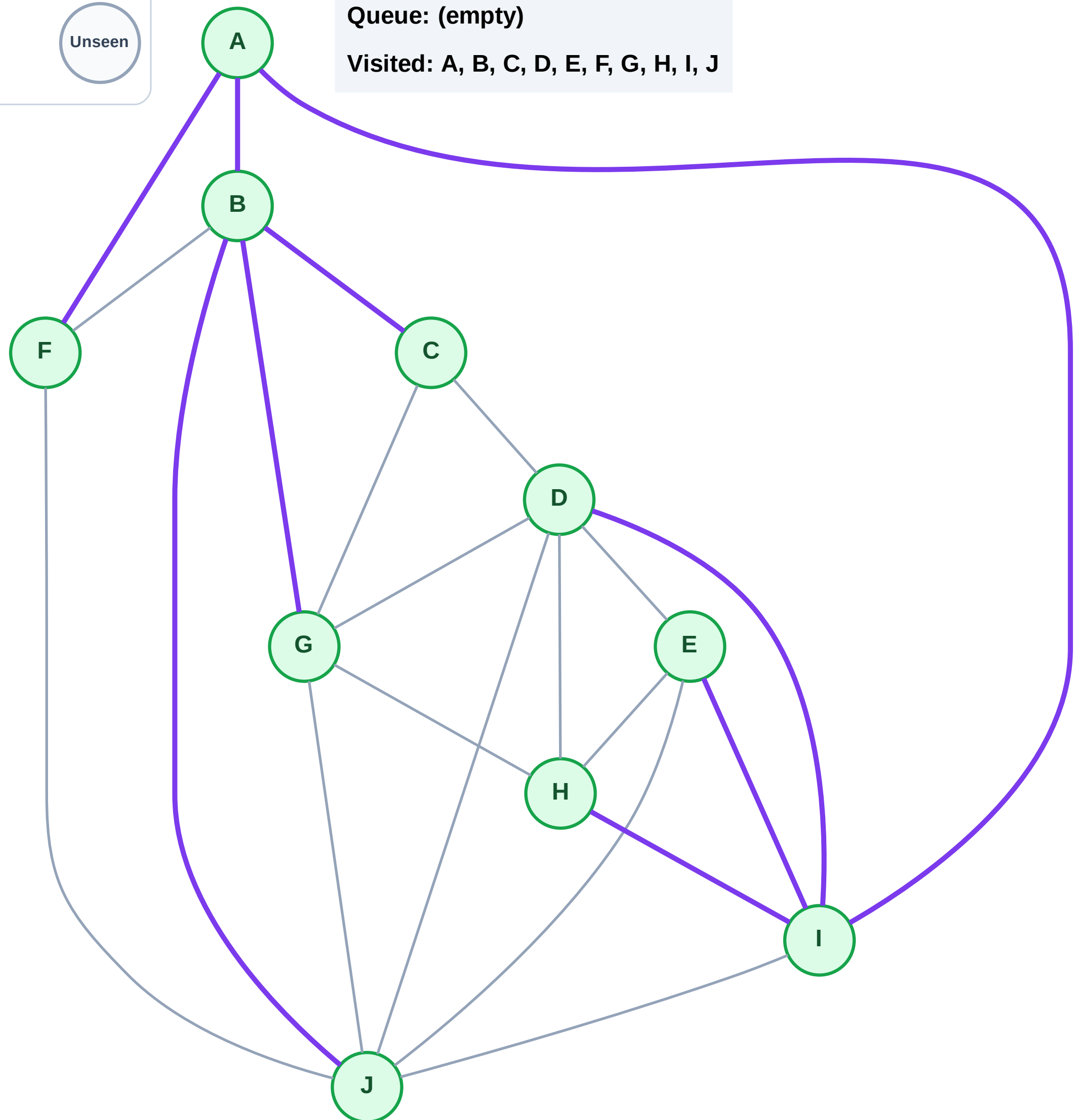
Step 21: No new neighbors from H

Legend



Queue: (empty)

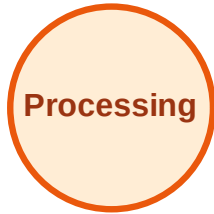
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

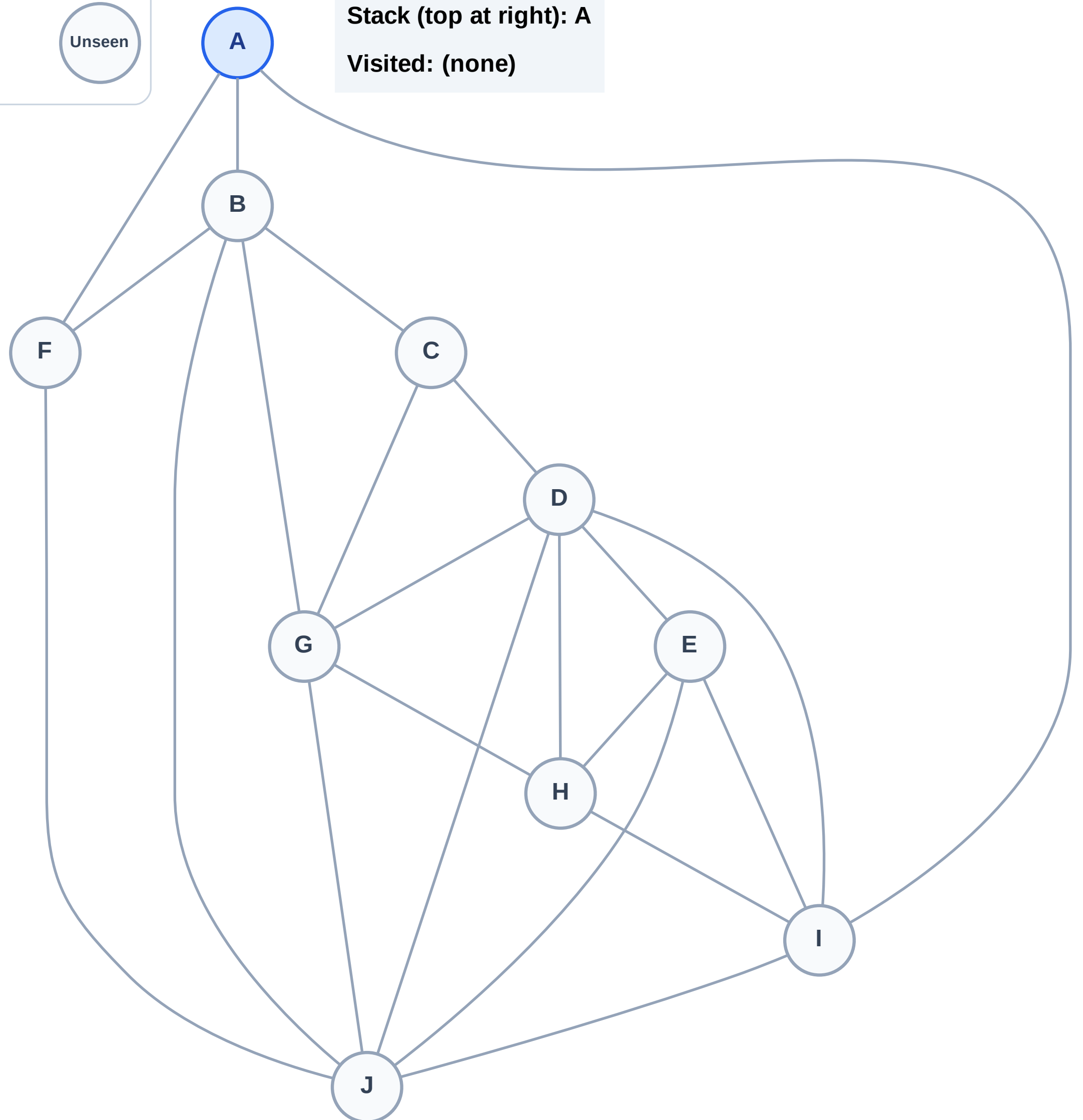
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

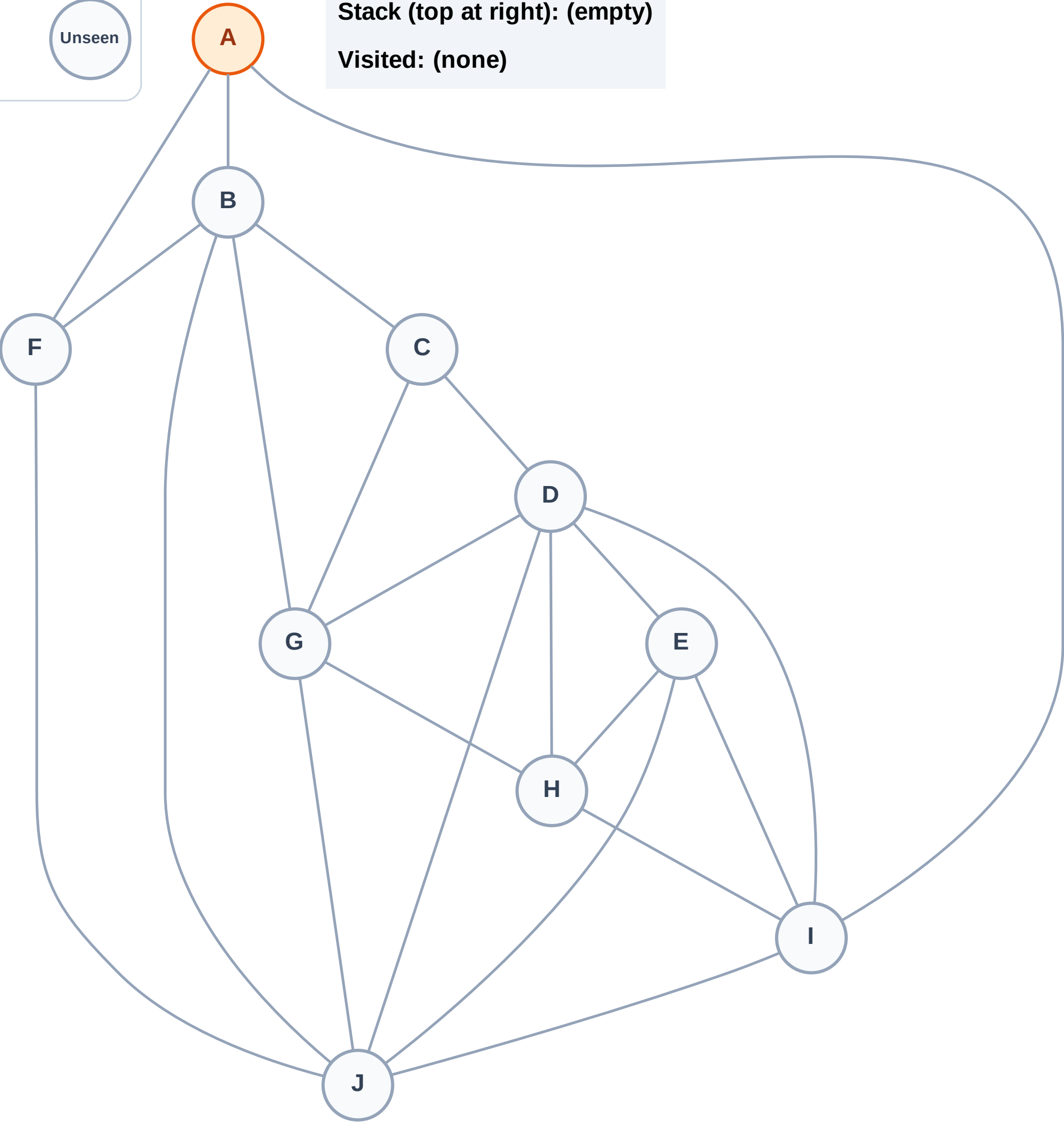
Complete

Processing

Discovered

Unseen

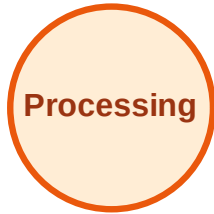
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

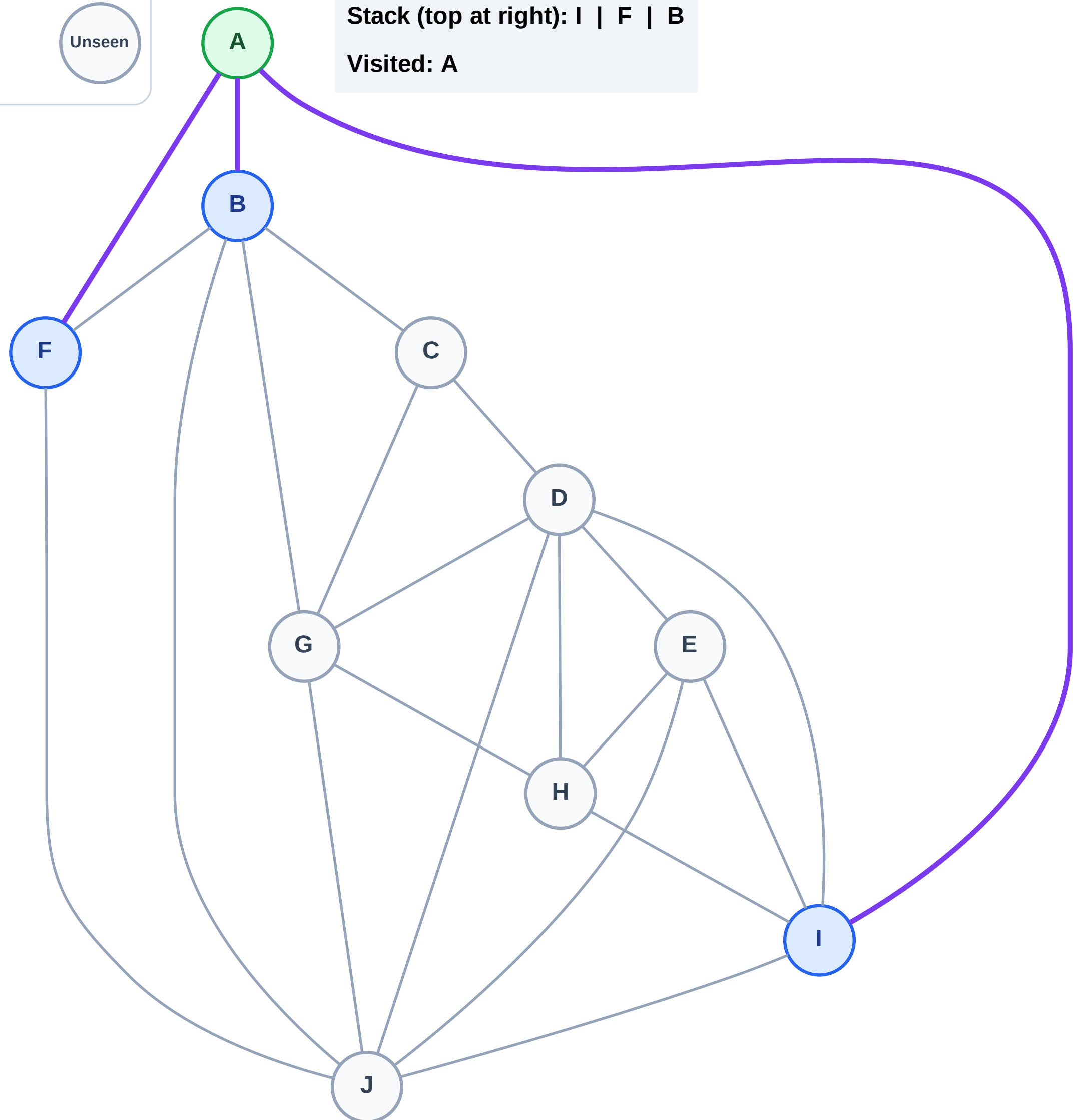
Step 3: Discover I, F, B from A

Legend



Stack (top at right): I | F | B

Visited: A



DFS Traversal

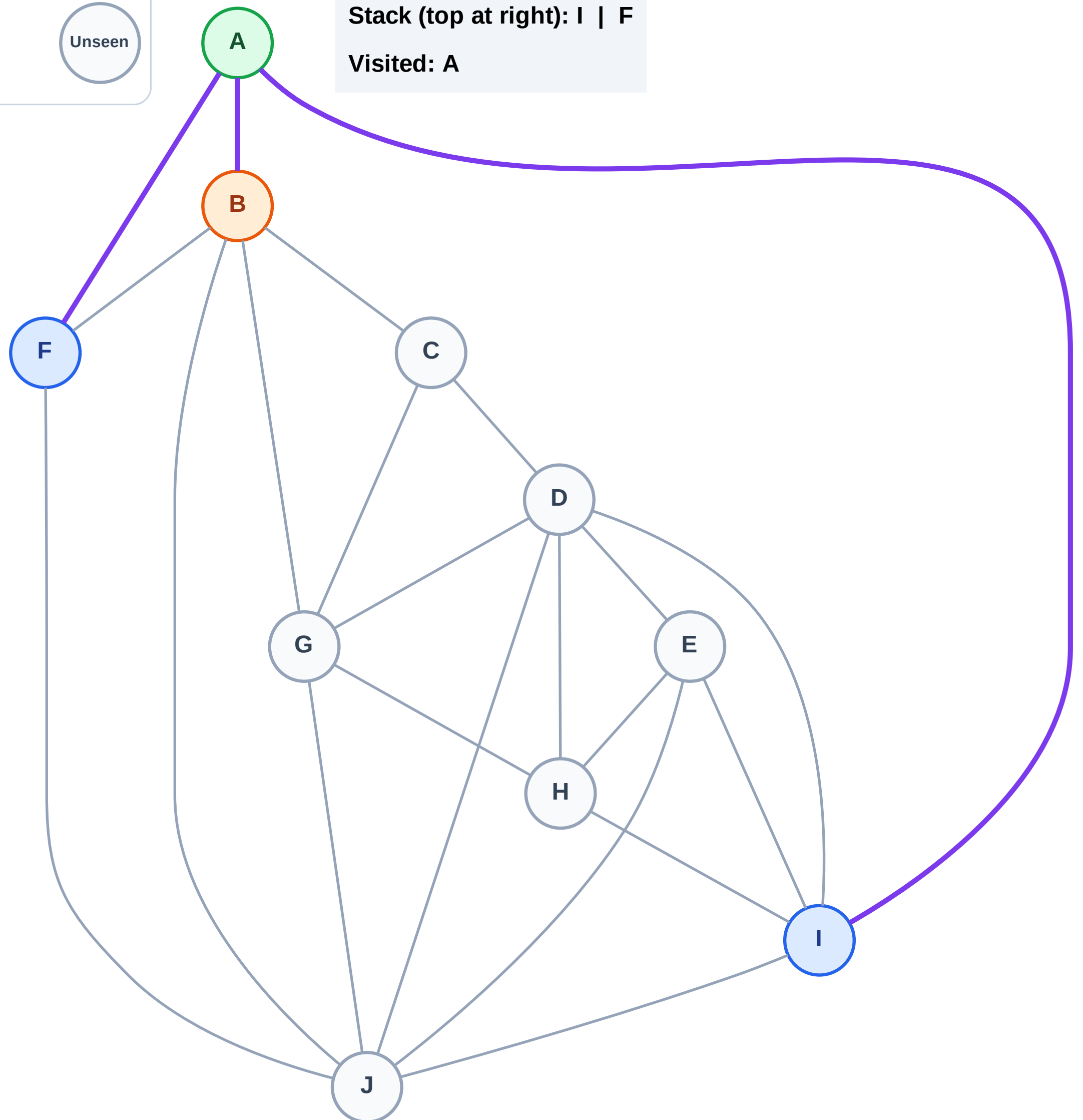
Step 4: Process node B

Legend



Stack (top at right): I | F

Visited: A



DFS Traversal

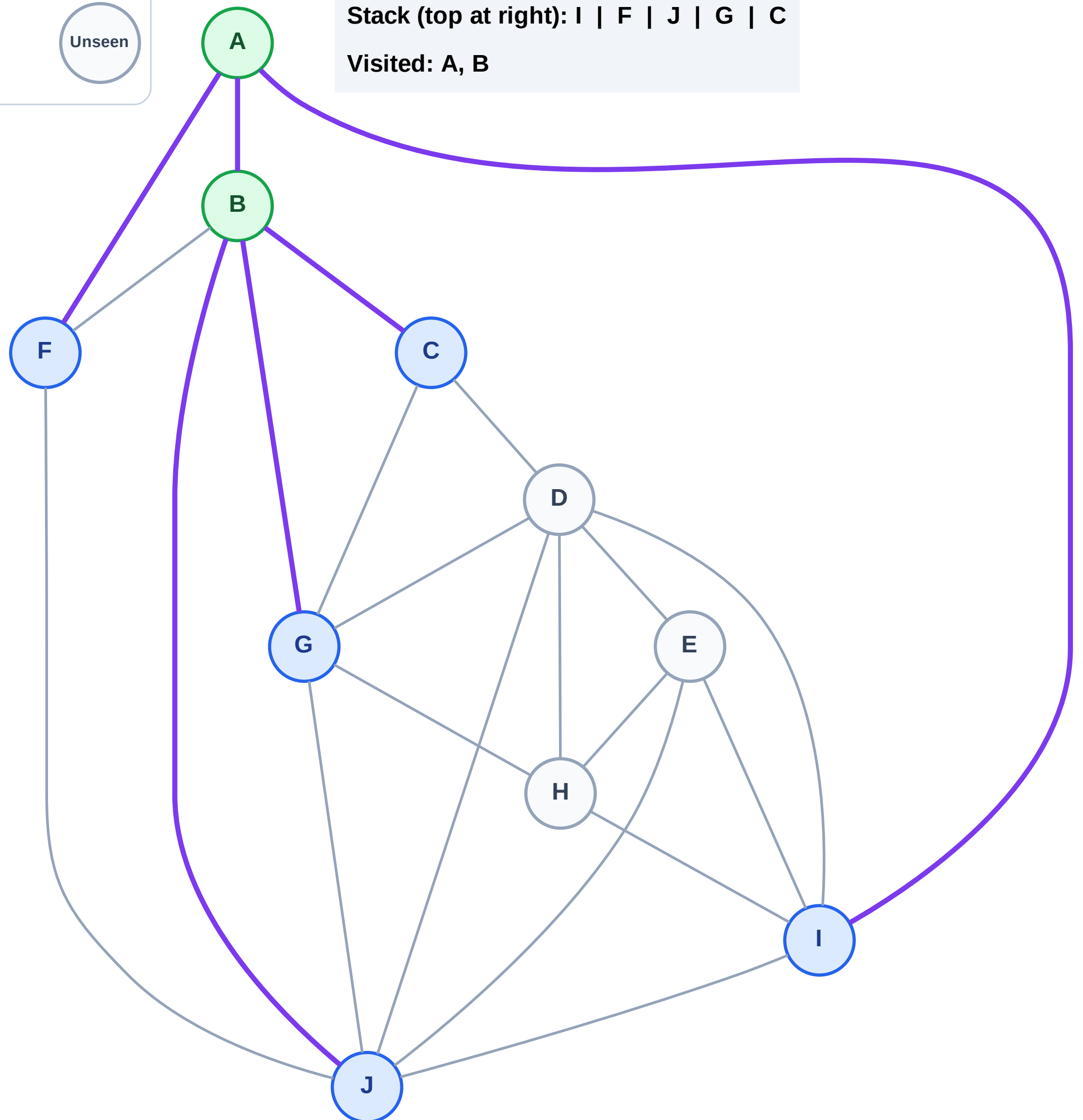
Step 5: Discover J, G, C from B

Legend



Stack (top at right): I | F | J | G | C

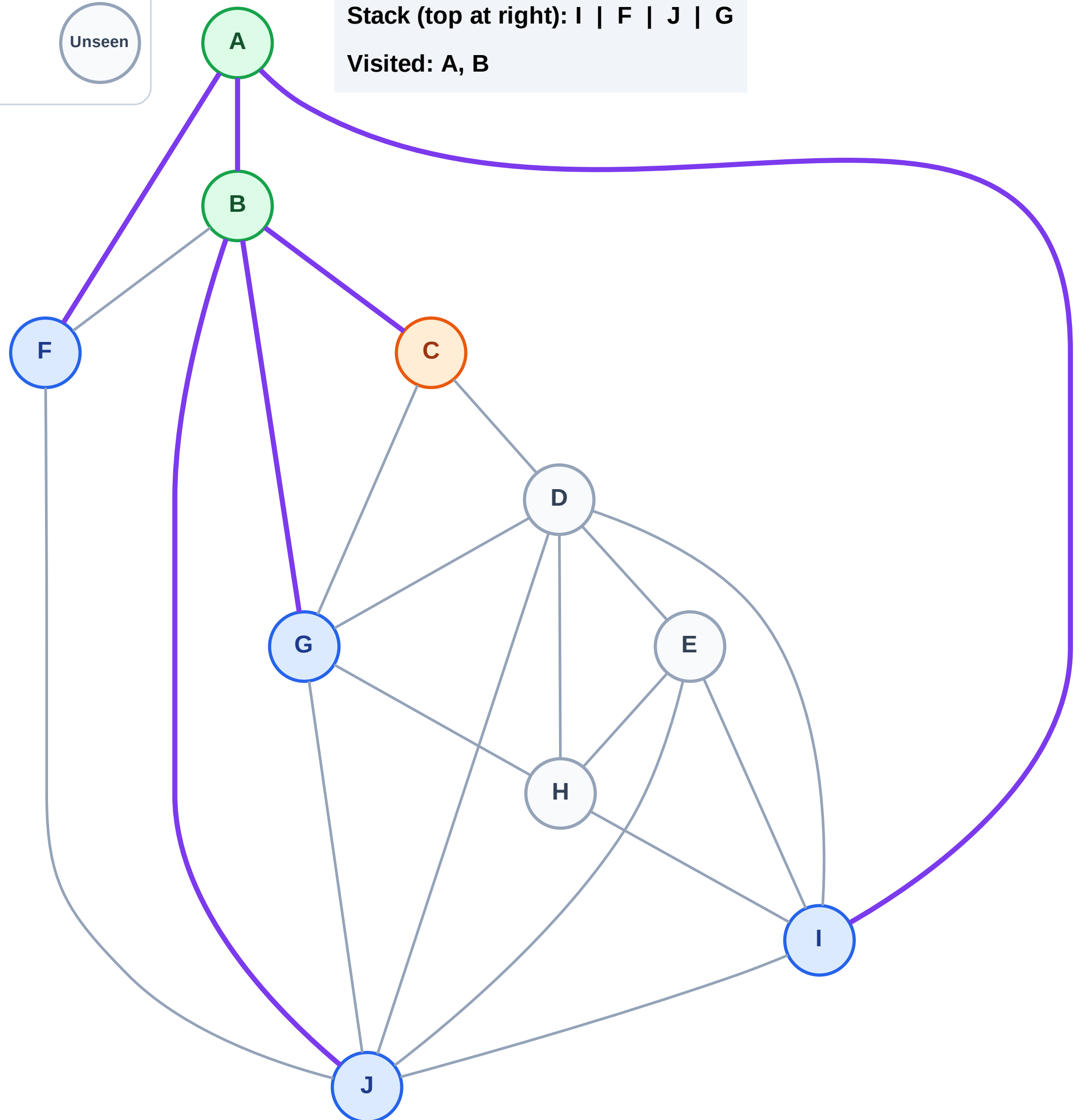
Visited: A, B



Step 6: Process node C



Visited: A B



DFS Traversal

Step 7: Discover D from C

Legend

Complete

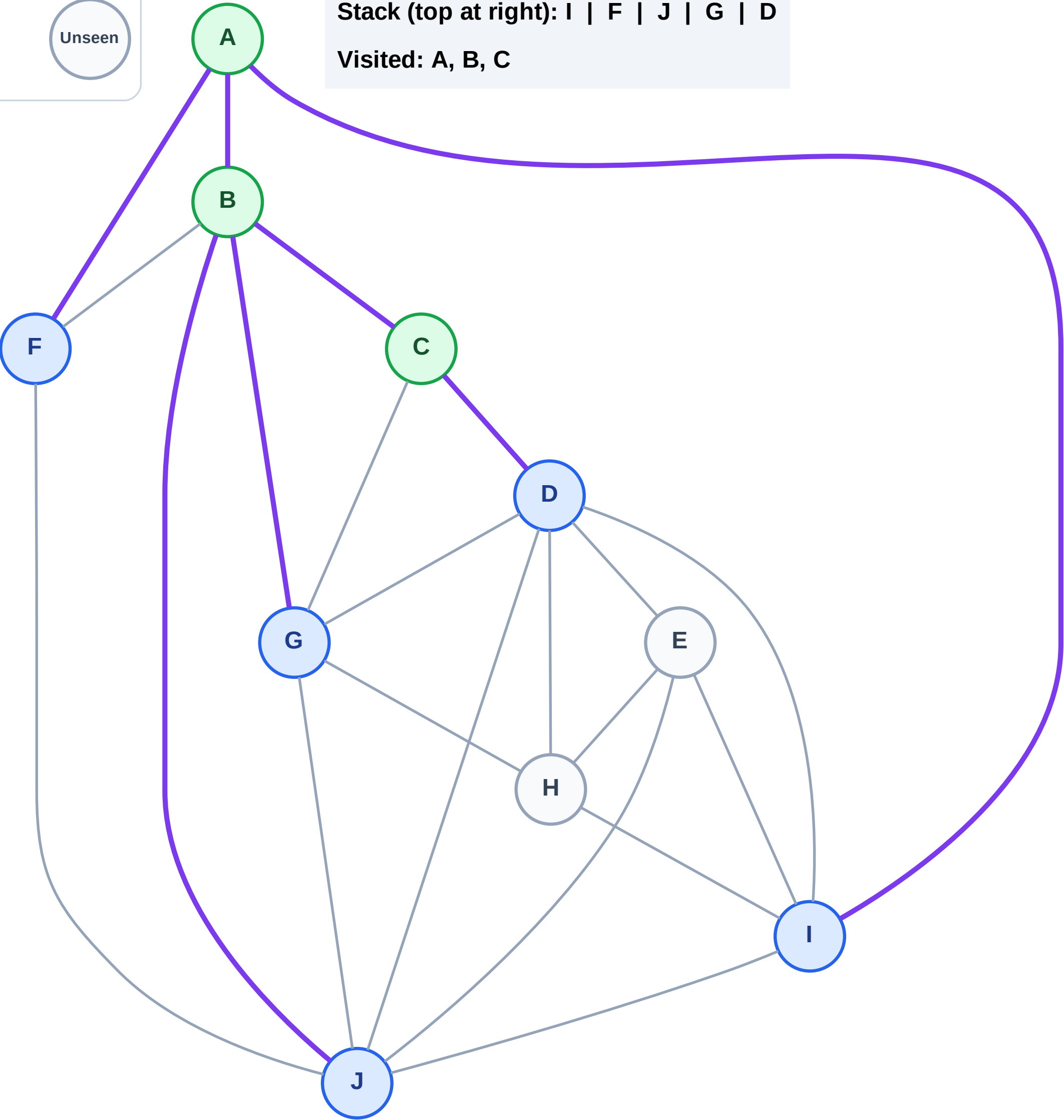
Processing

Discovered

Unseen

Stack (top at right): I | F | J | G | D

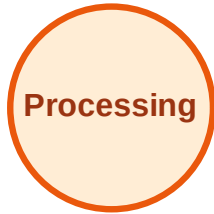
Visited: A, B, C



DFS Traversal

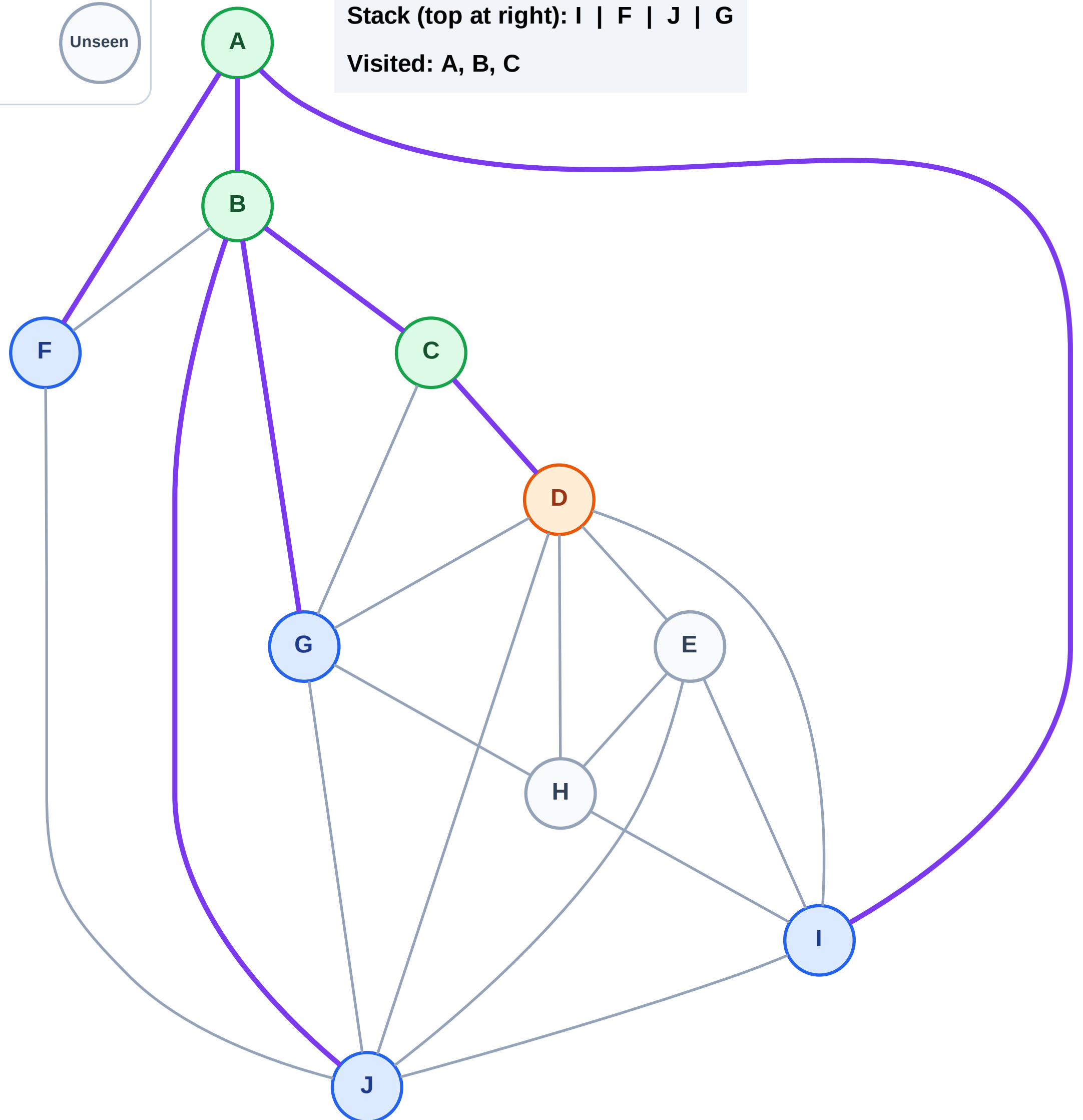
Step 8: Process node D

Legend



Stack (top at right): I | F | J | G

Visited: A, B, C



DFS Traversal

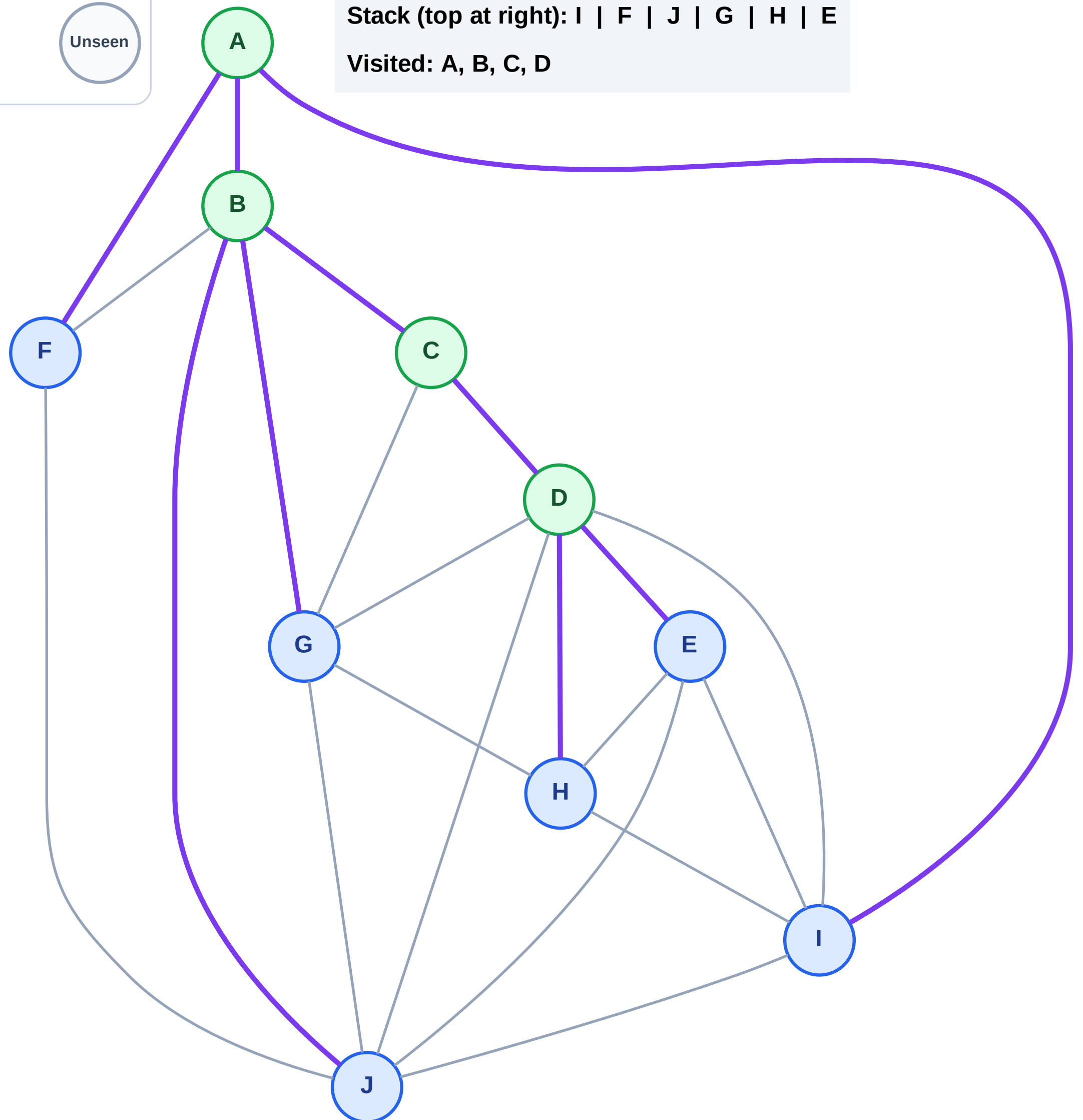
Step 9: Discover H, E from D

Legend



Stack (top at right): I | F | J | G | H | E

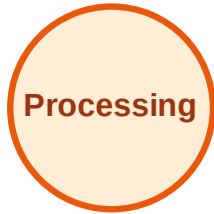
Visited: A, B, C, D



DFS Traversal

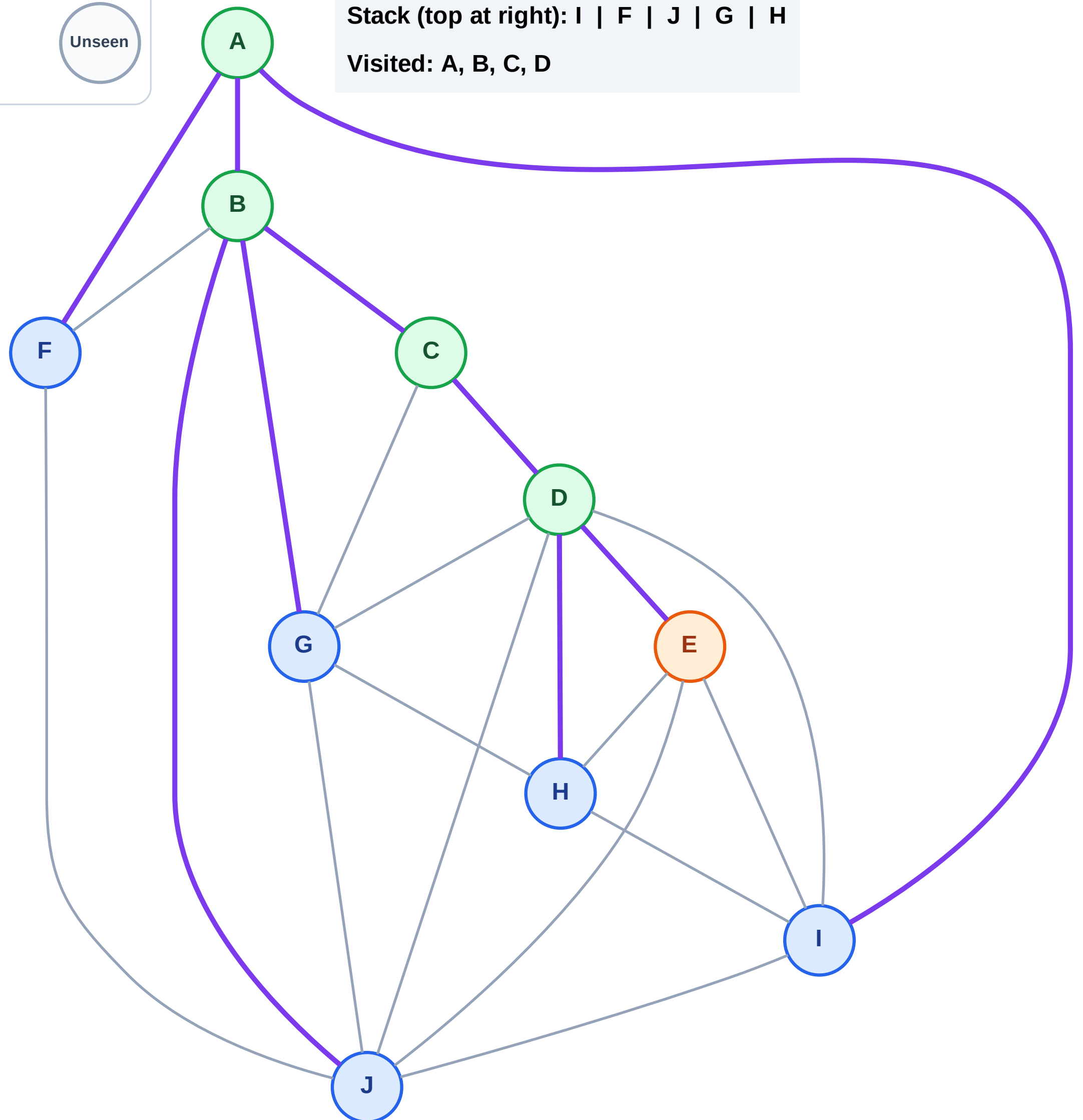
Step 10: Process node E

Legend



Stack (top at right): I | F | J | G | H

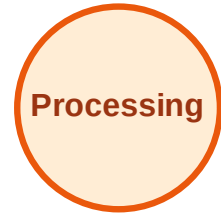
Visited: A, B, C, D



DFS Traversal

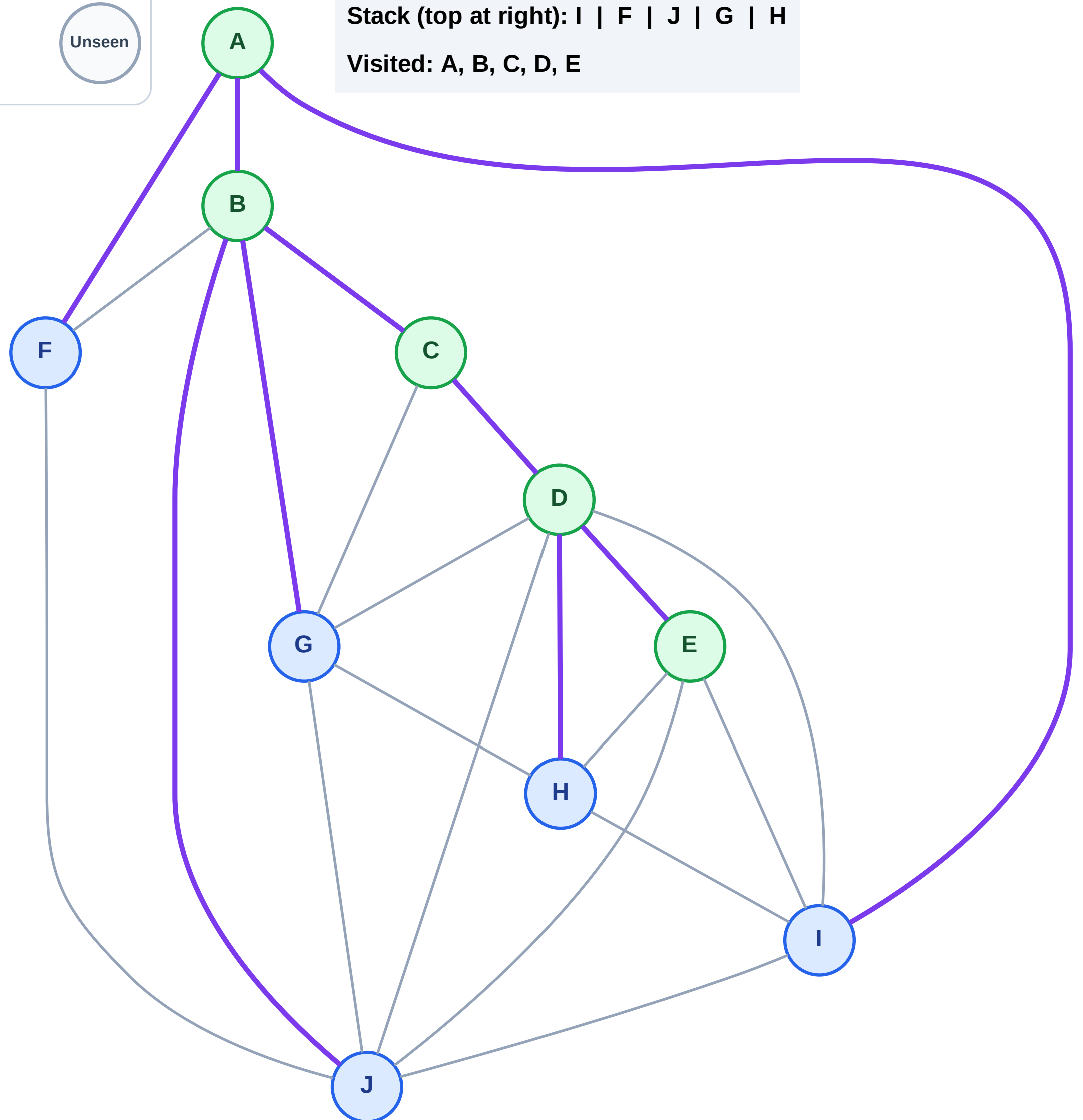
Step 11: No new neighbors from E

Legend



Stack (top at right): I | F | J | G | H

Visited: A, B, C, D, E

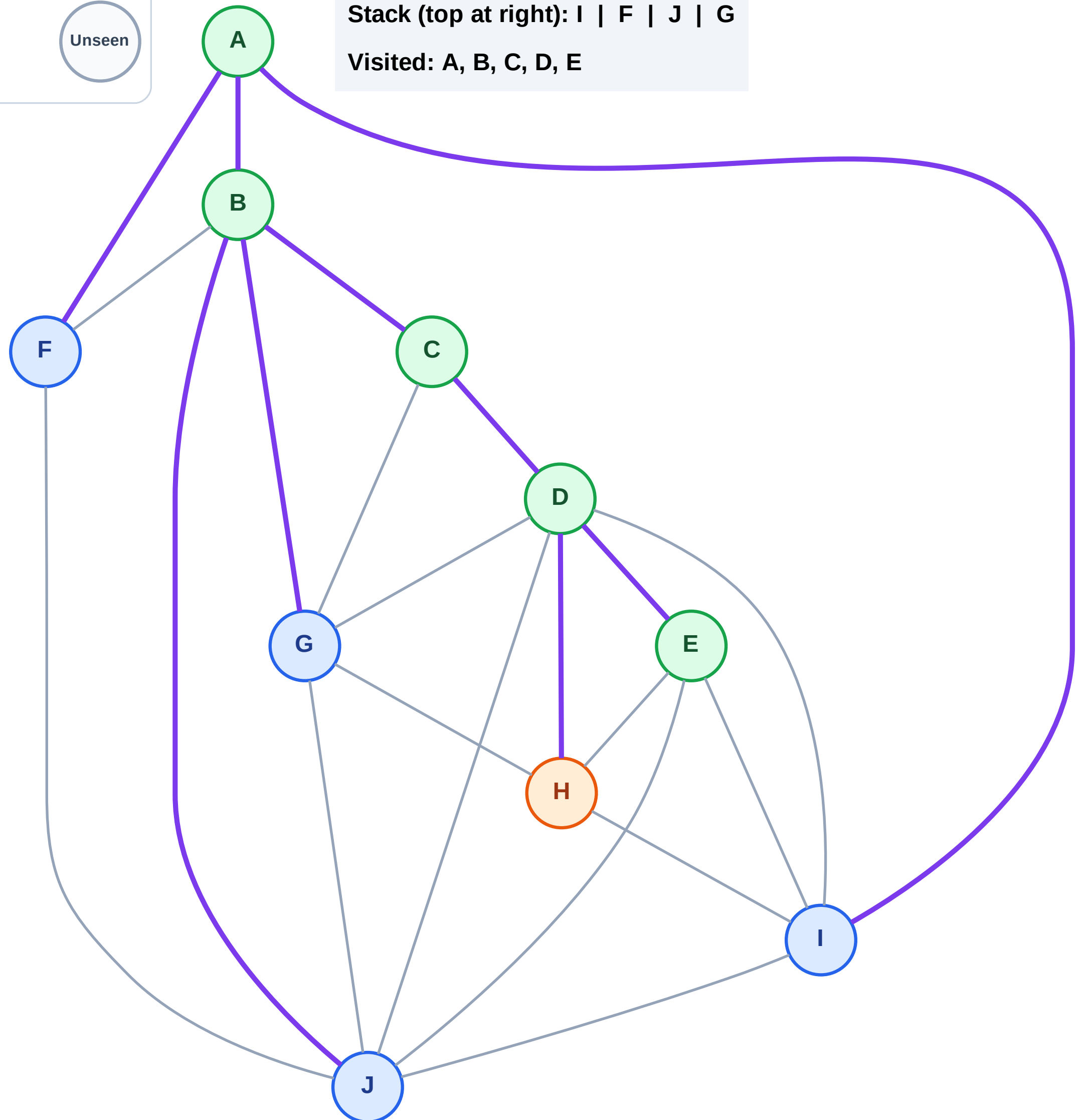


Step 12: Process node H

Legend

Unseen

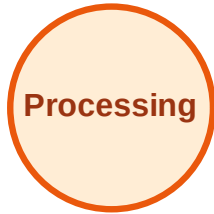
Visited: A, B, C, D, E



DFS Traversal

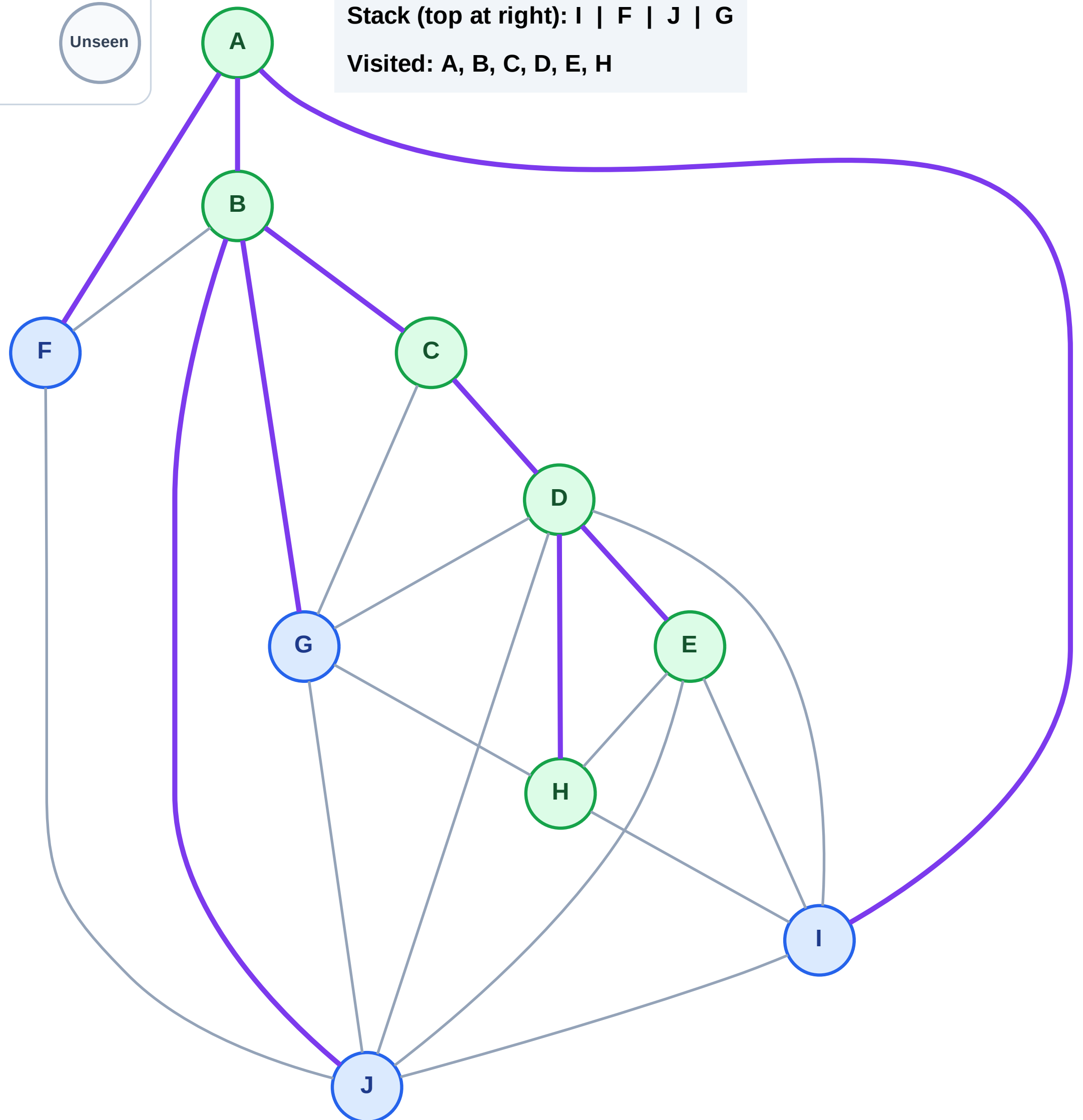
Step 13: No new neighbors from H

Legend



Stack (top at right): I | F | J | G

Visited: A, B, C, D, E, H



DFS Traversal

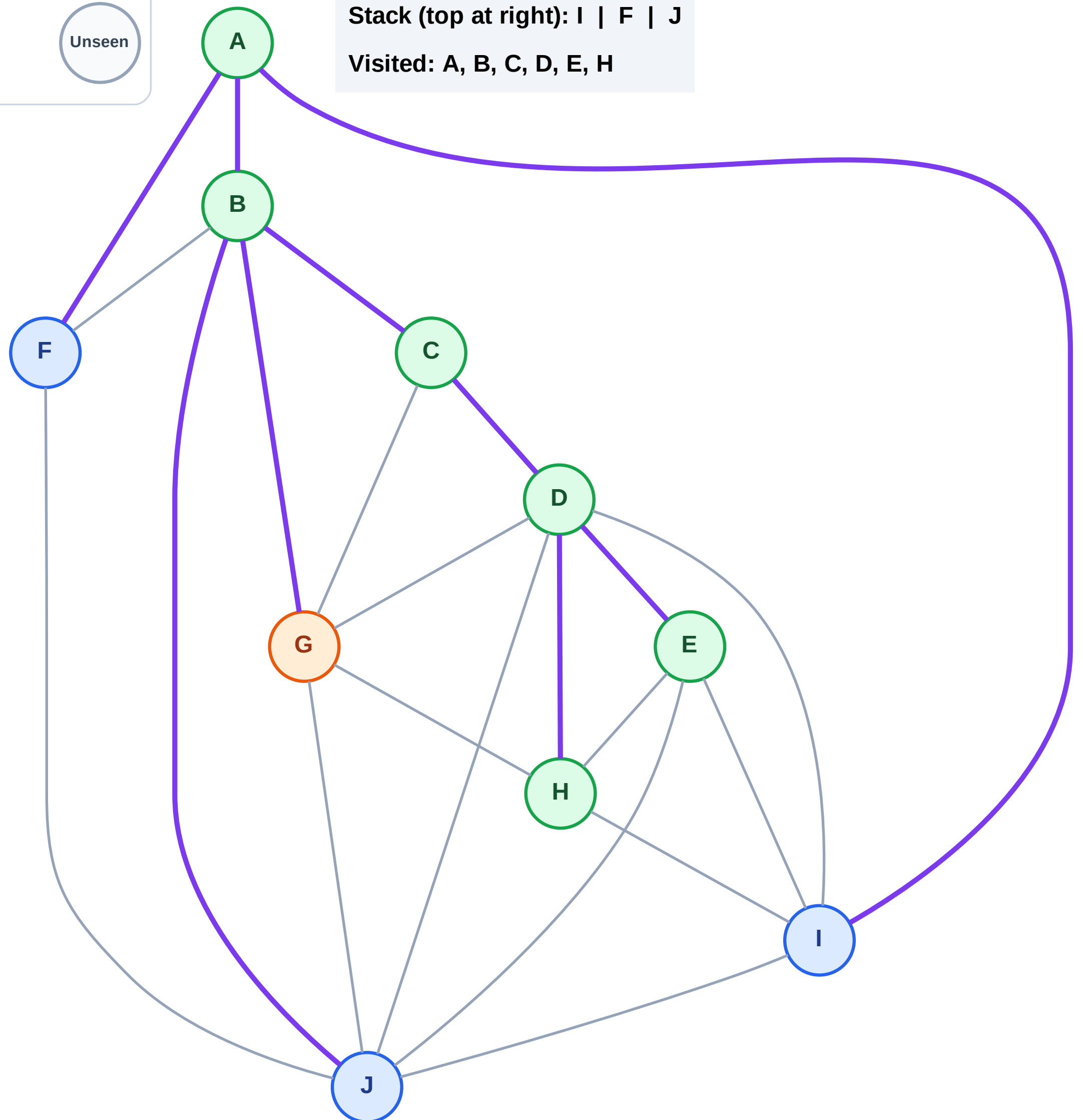
Step 14: Process node G

Legend



Stack (top at right): I | F | J

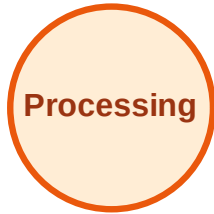
Visited: A, B, C, D, E, H



DFS Traversal

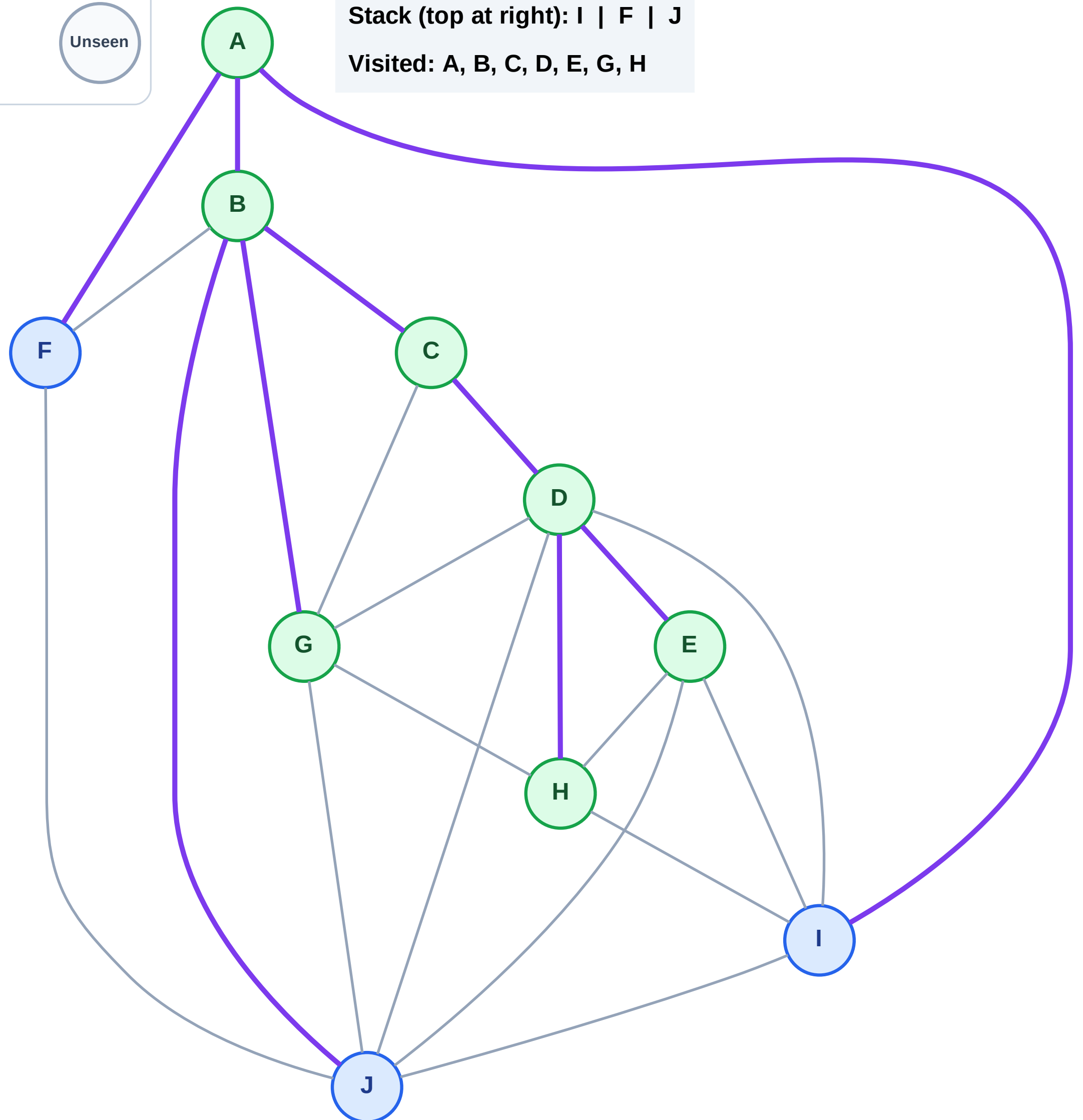
Step 15: No new neighbors from G

Legend

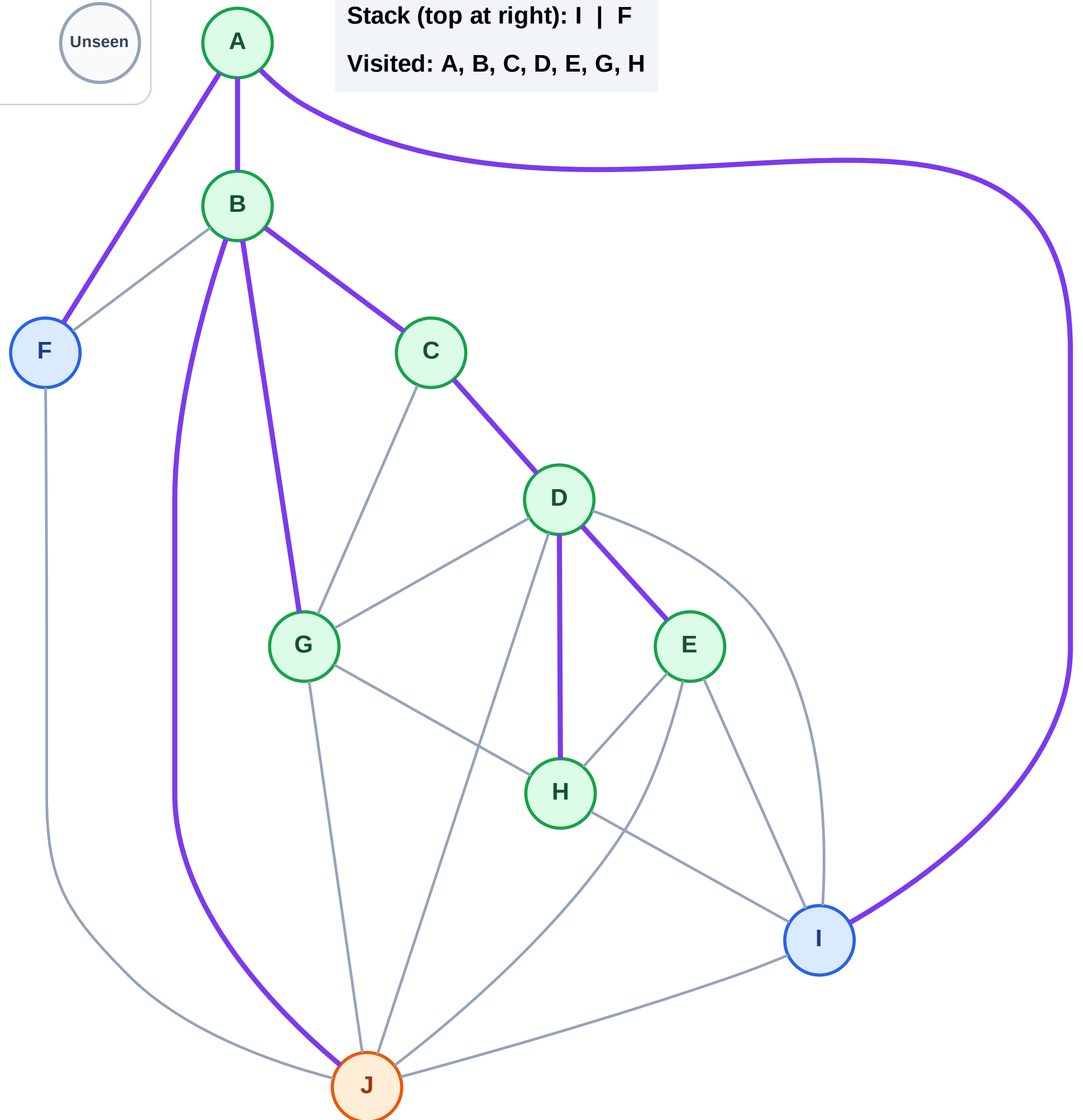
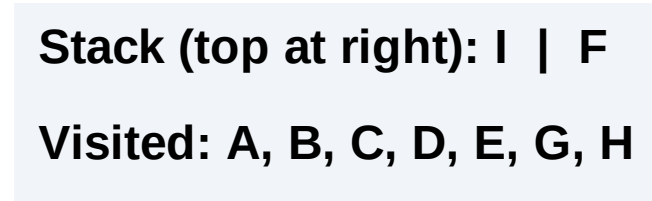


Stack (top at right): I | F | J

Visited: A, B, C, D, E, G, H



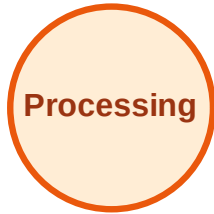
Step 16: Process node J



DFS Traversal

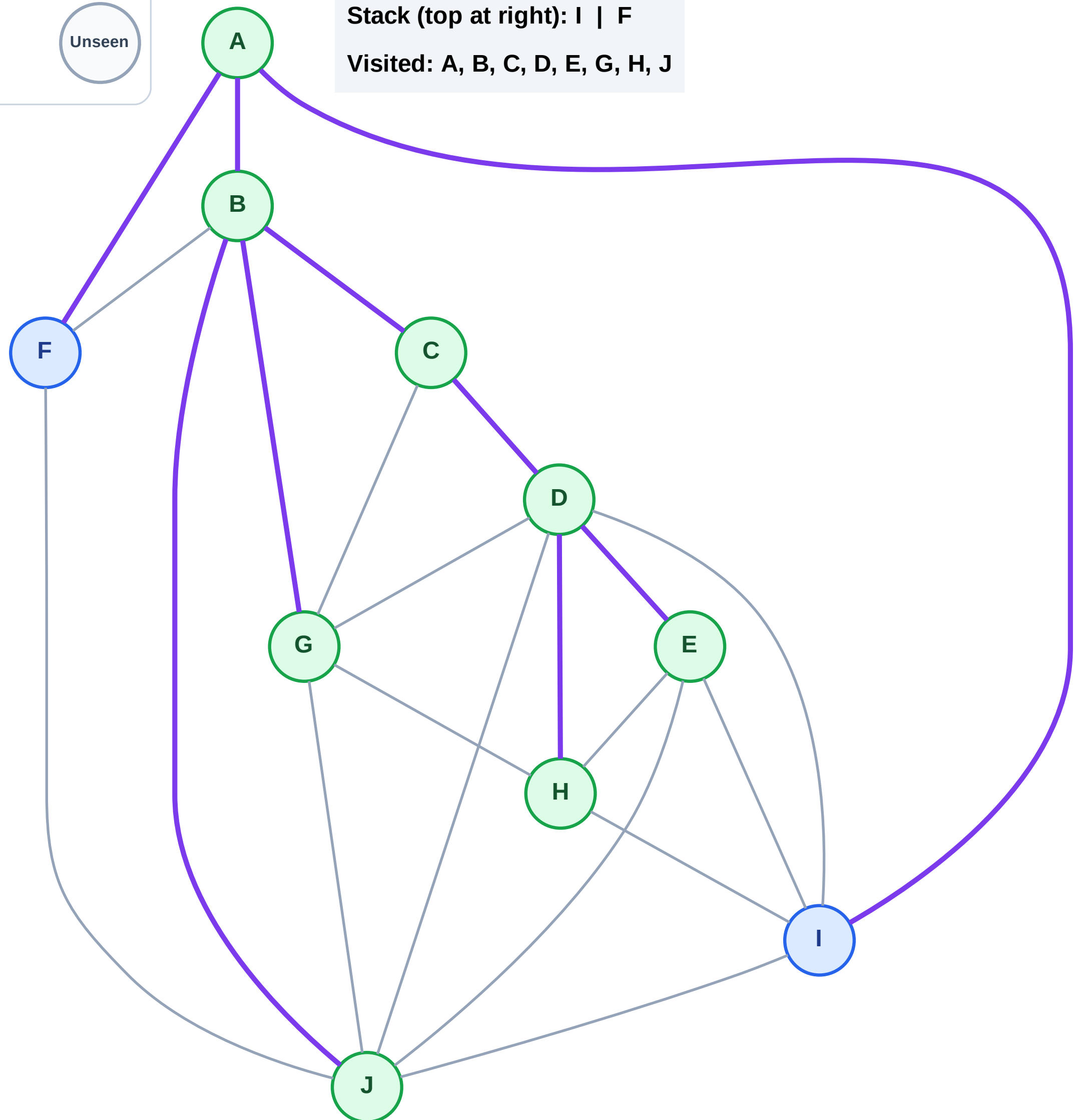
Step 17: No new neighbors from J

Legend



Stack (top at right): I | F

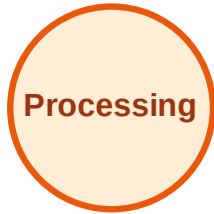
Visited: A, B, C, D, E, G, H, J



DFS Traversal

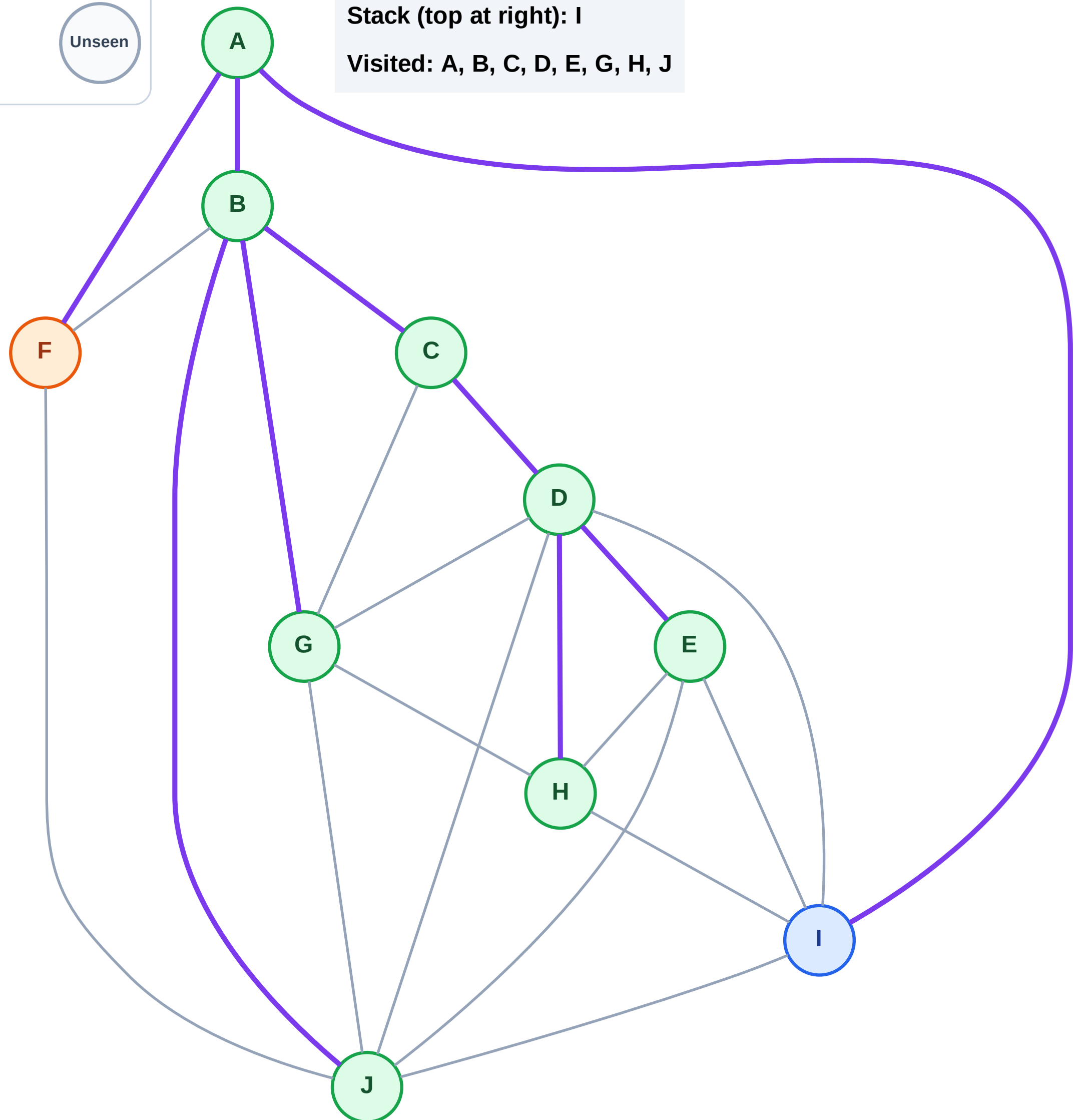
Step 18: Process node F

Legend



Stack (top at right): I

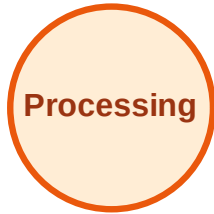
Visited: A, B, C, D, E, G, H, J



DFS Traversal

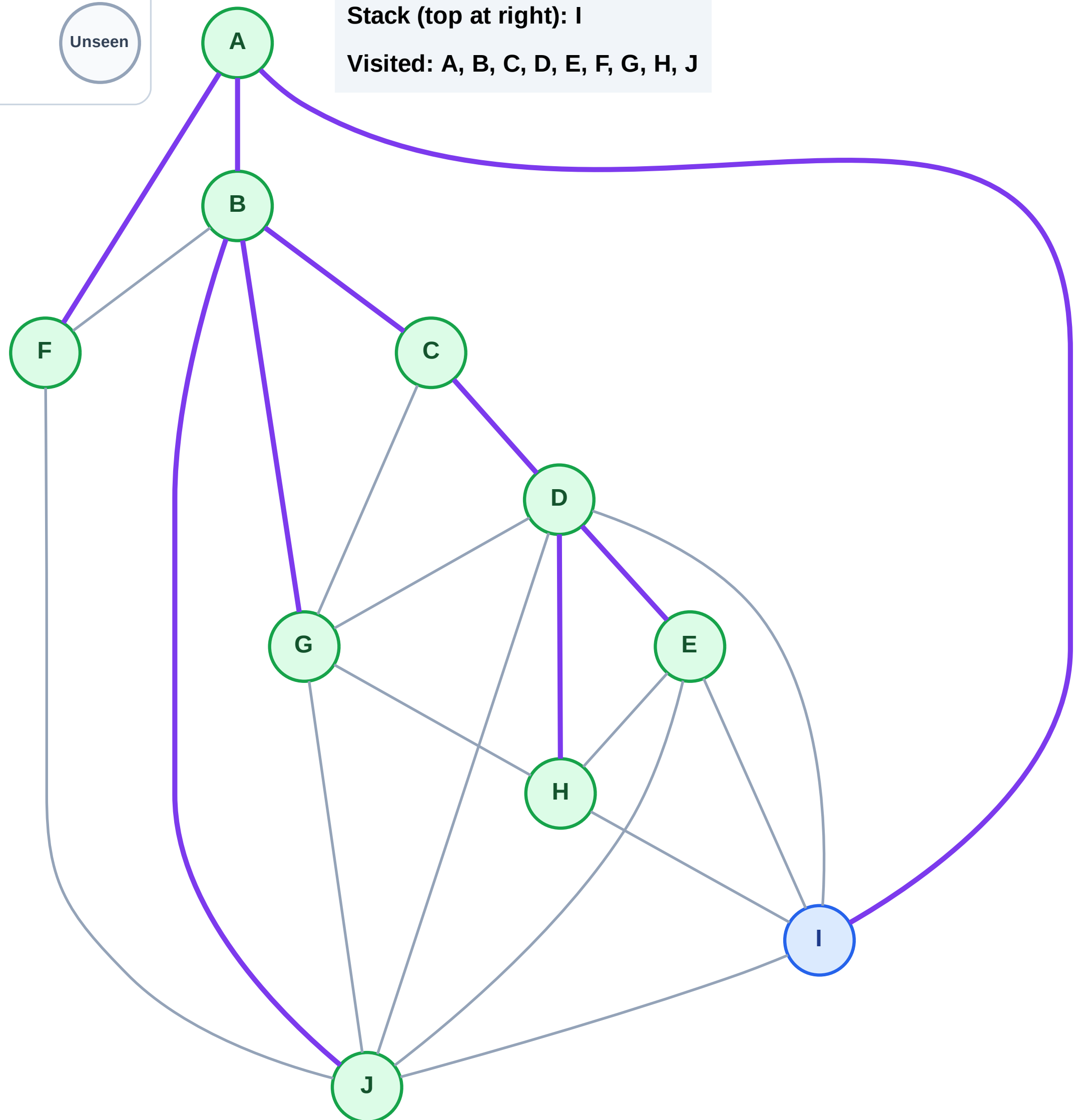
Step 19: No new neighbors from F

Legend



Stack (top at right): I

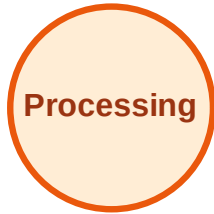
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

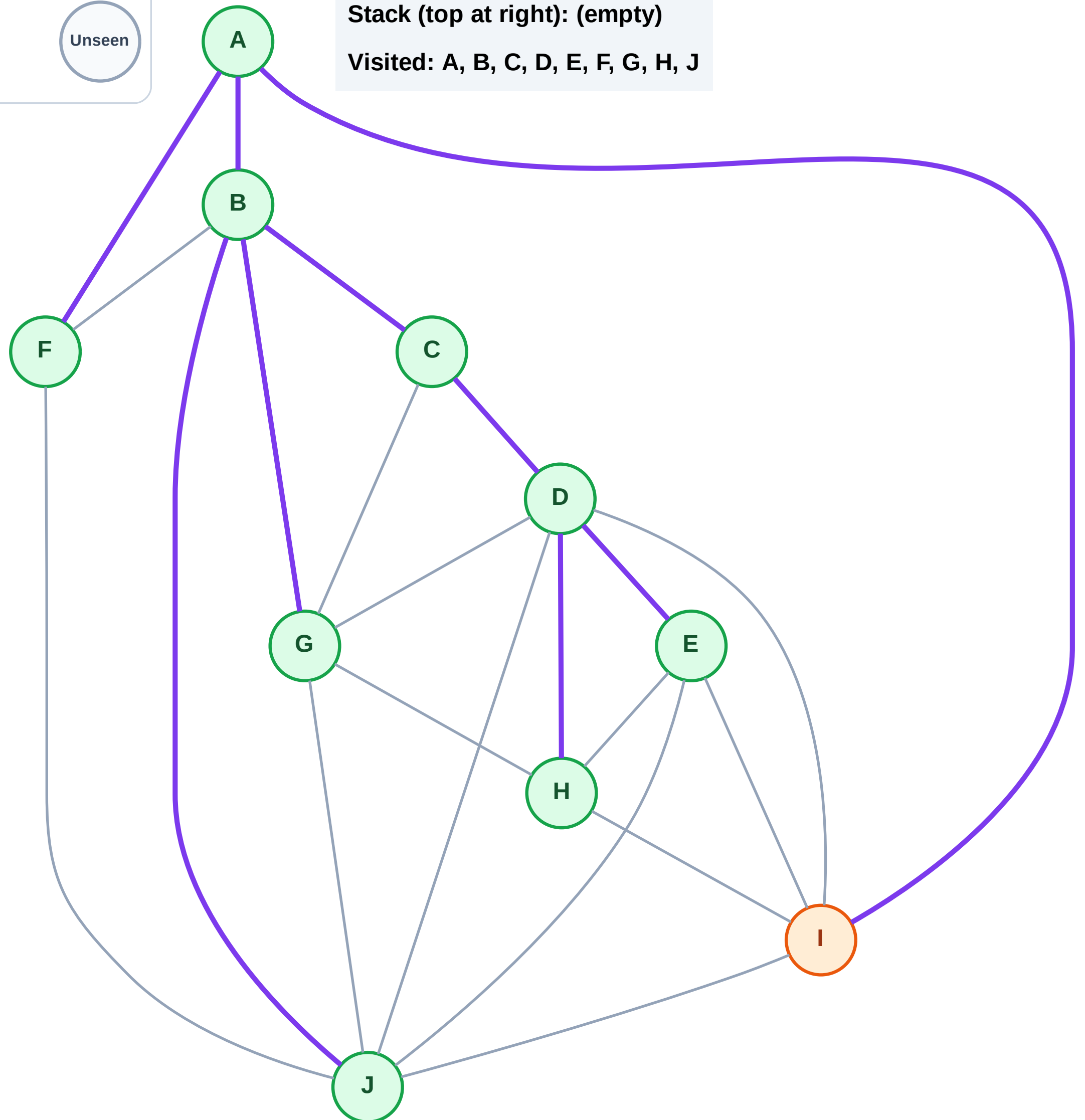
Step 20: Process node I

Legend



Stack (top at right): (empty)

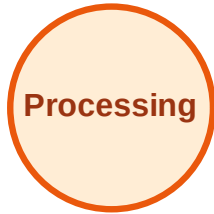
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

